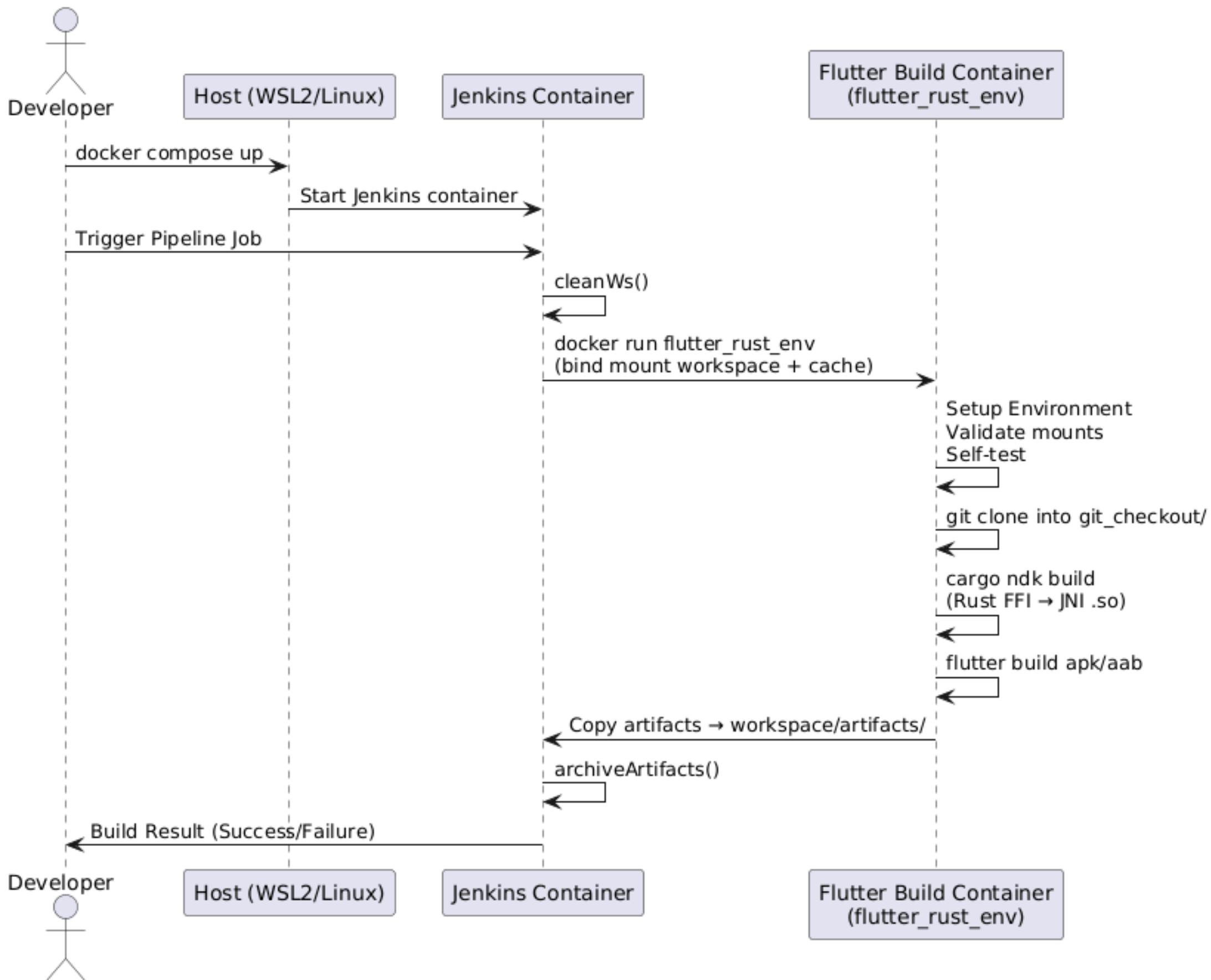
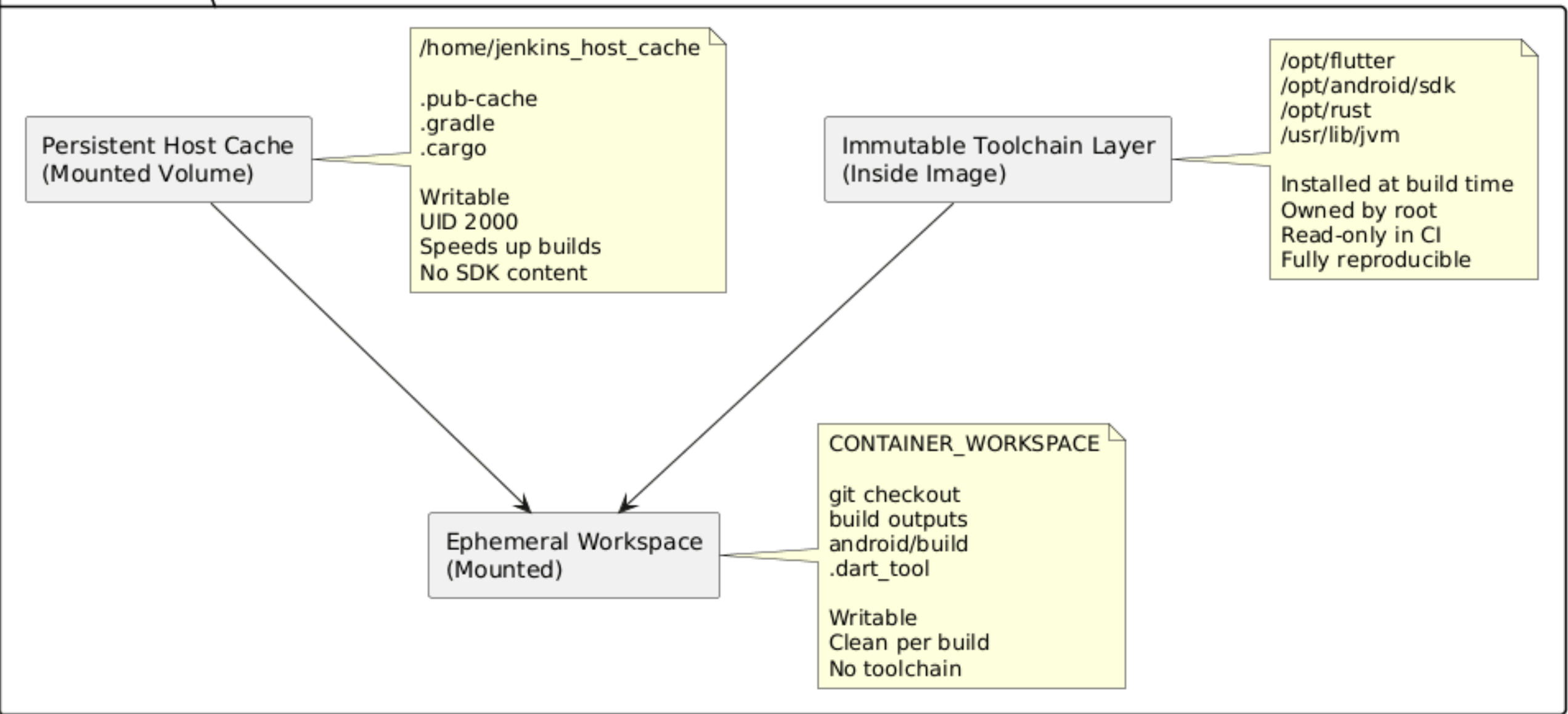


Deployment Sequence - Host → Jenkins → Flutter Container

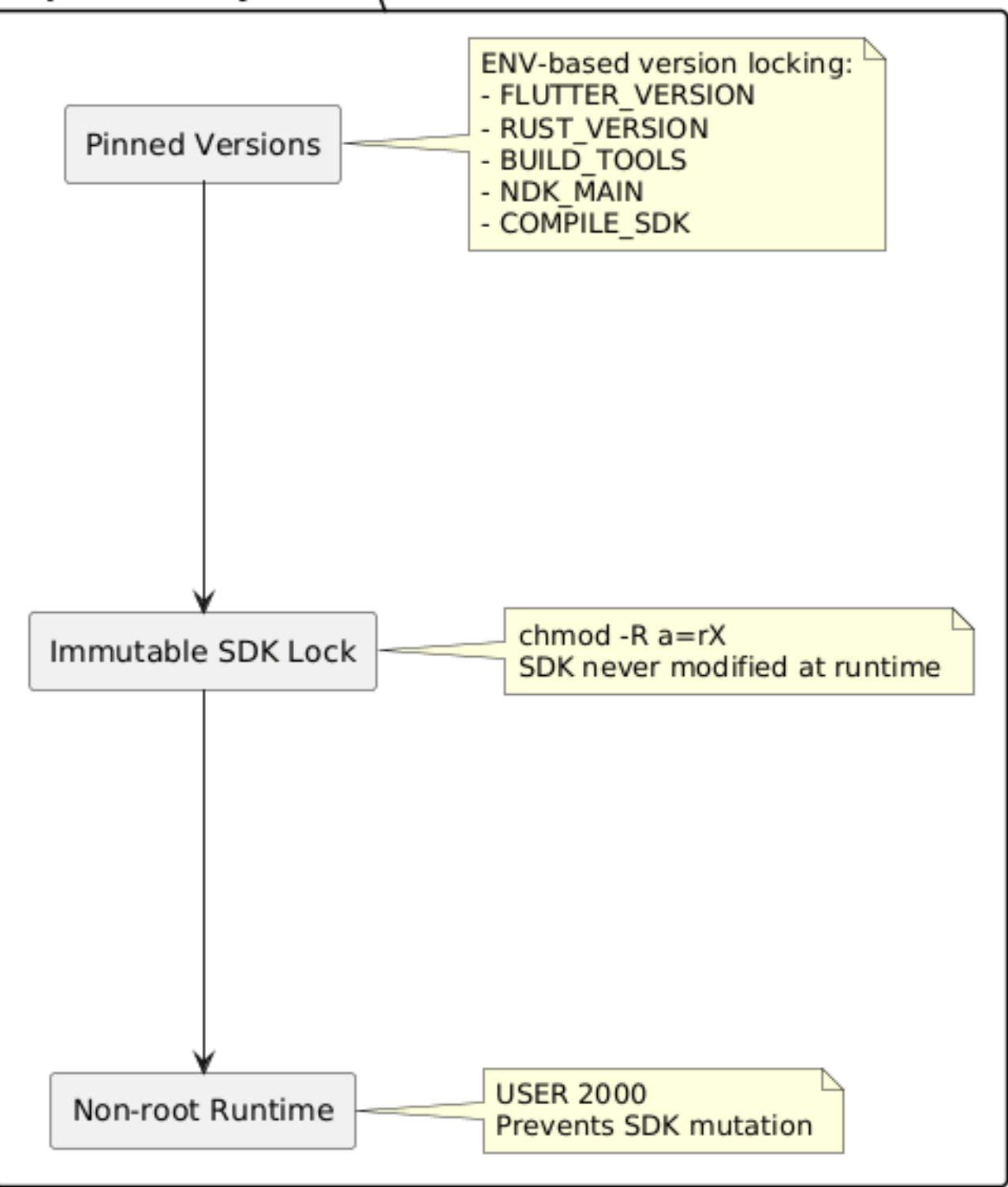


Immutable Toolchain Versions	
Java	: 17
Flutter	: 3.35.7
Rust	: 1.91.1
Android SDK Tools	: 13114758
Android Compile	: 36
Android BuildTools	: 35.0.0
NDK	: 28.2.13676358
CMake	: 3.22.1
Base Image	: ubuntu 22.04

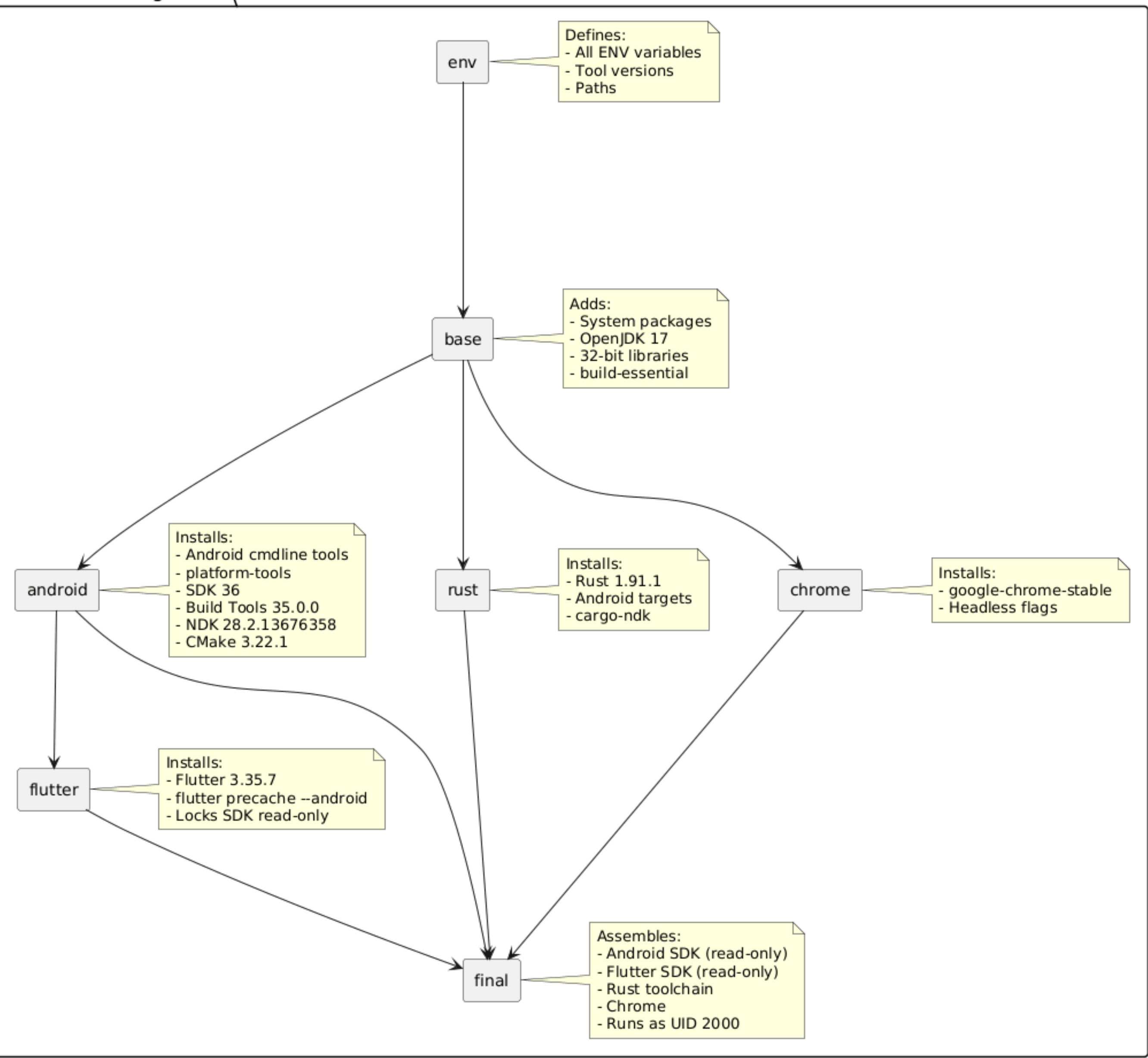
Security Model



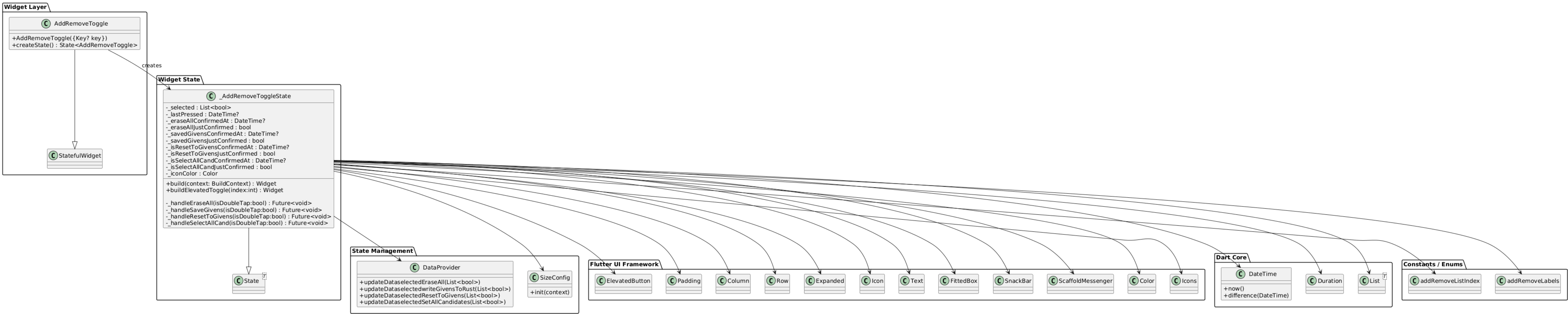
Reproducibility Model



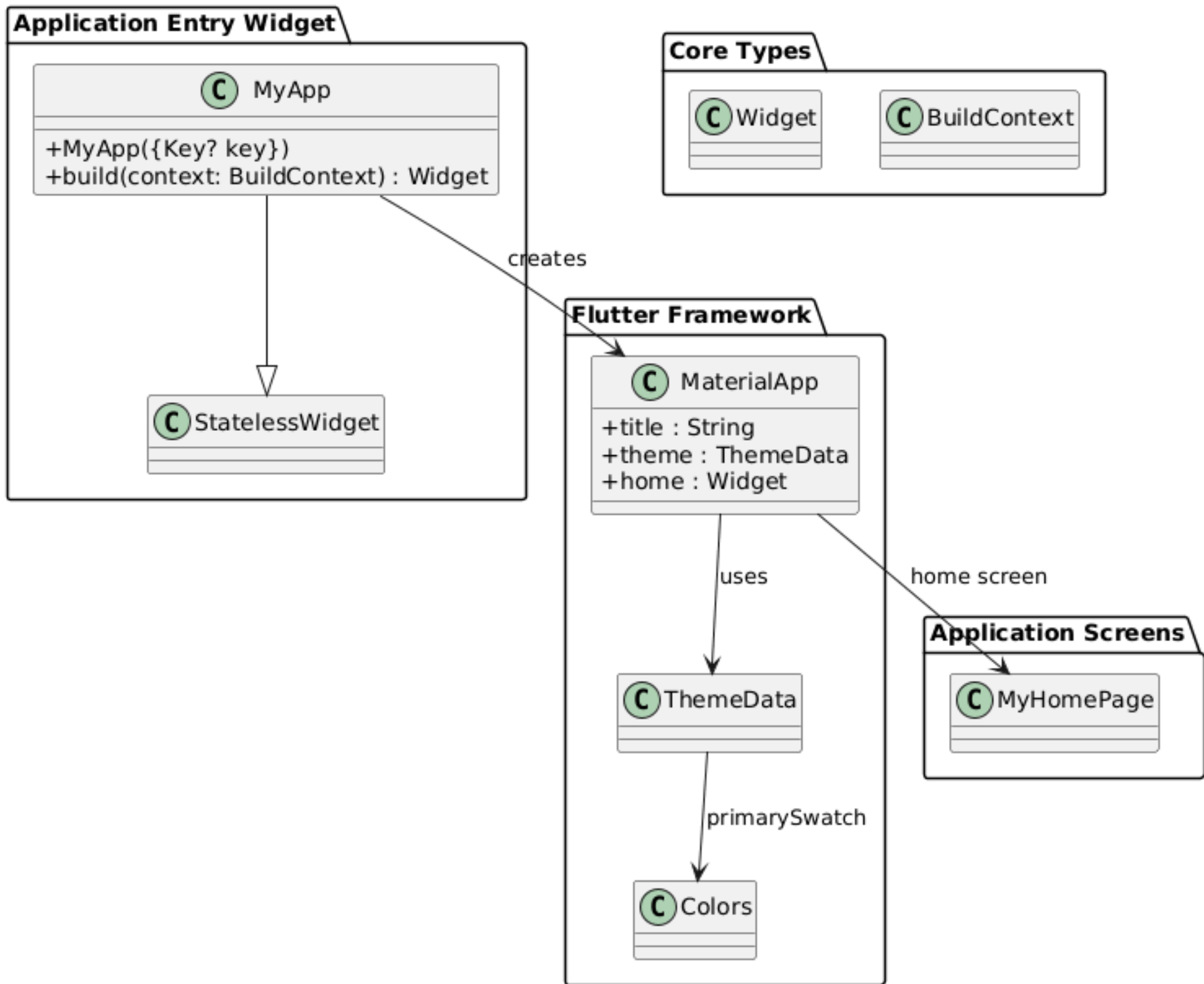
Docker Multi-Stage Build



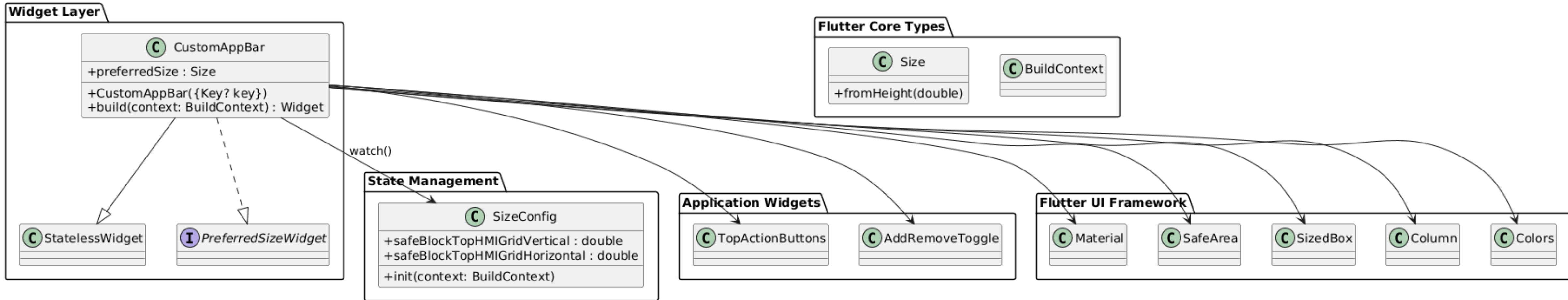
Flutter Widget Class Diagram - AddRemoveToggle



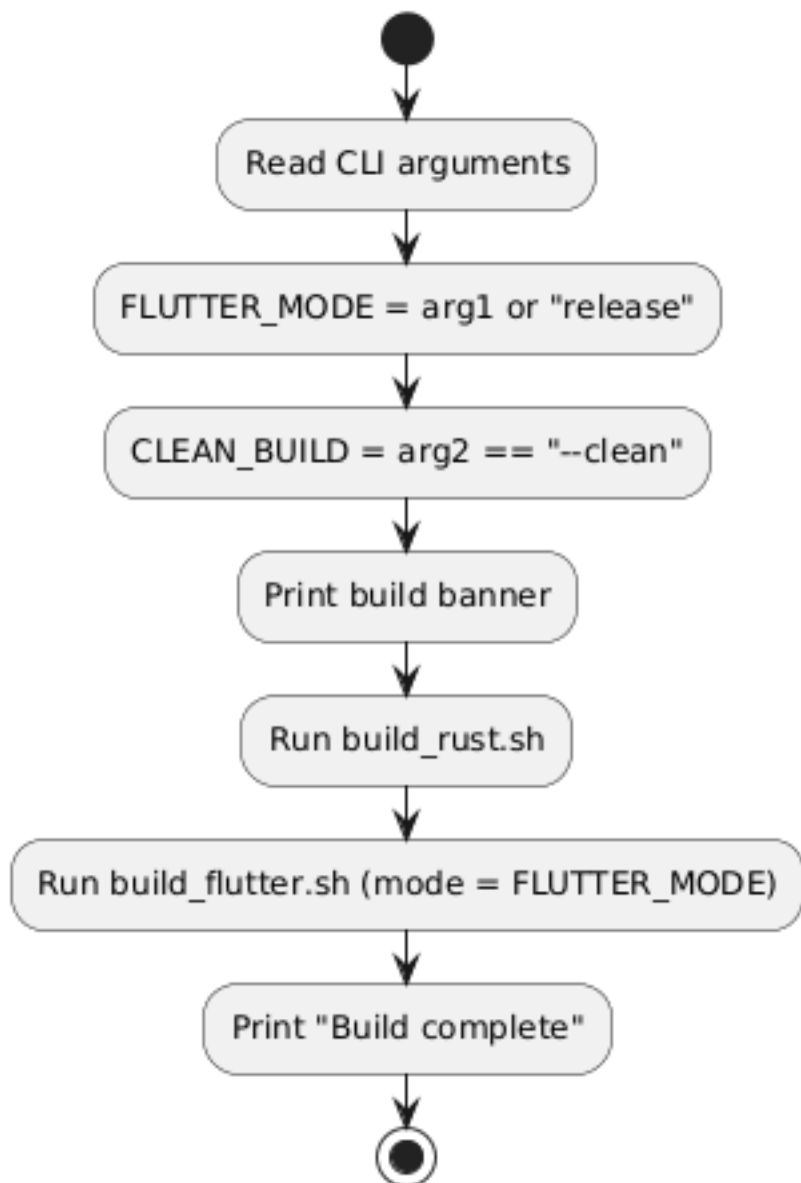
Flutter Widget Class Diagram - MyApp



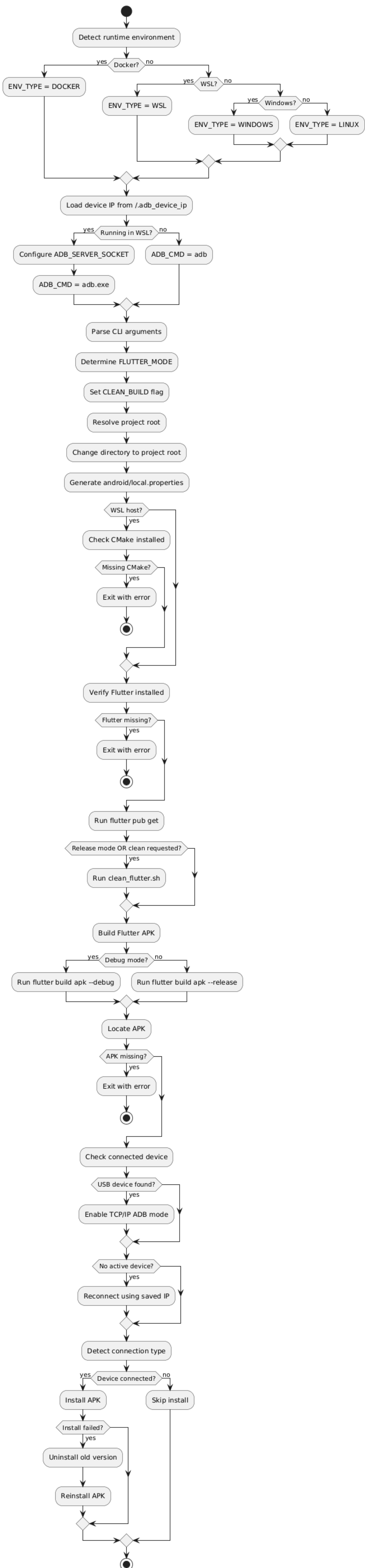
Flutter Widget Class Diagram - CustomAppBar



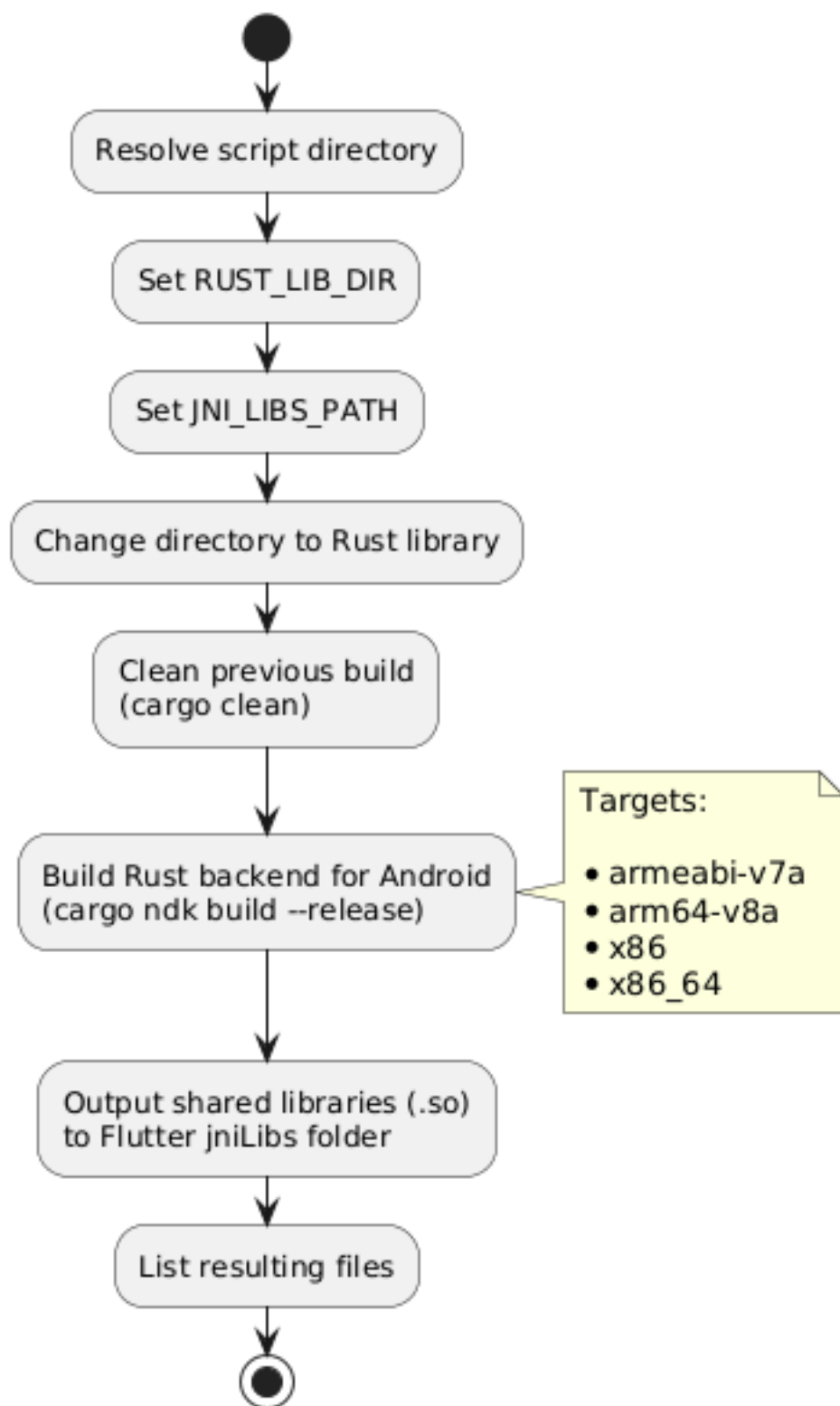
Rust + Flutter Build Pipeline



Flutter + Rust Build & Deploy Script



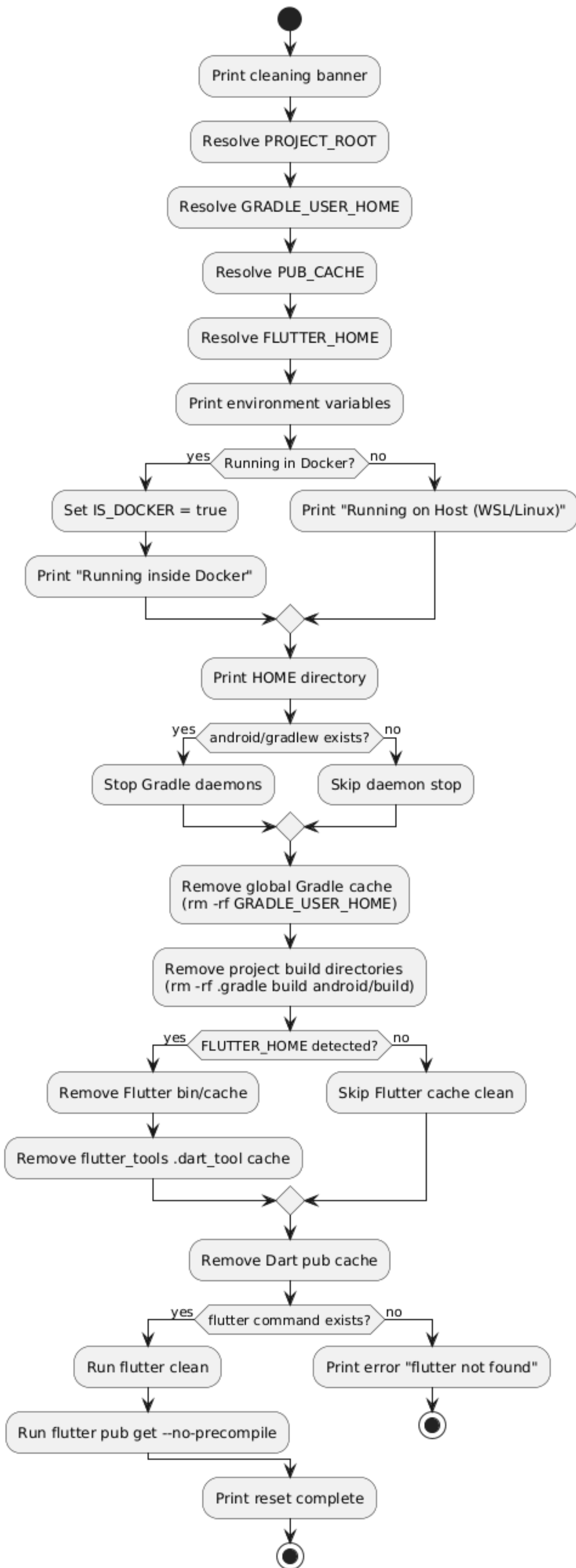
Rust Android Build Pipeline



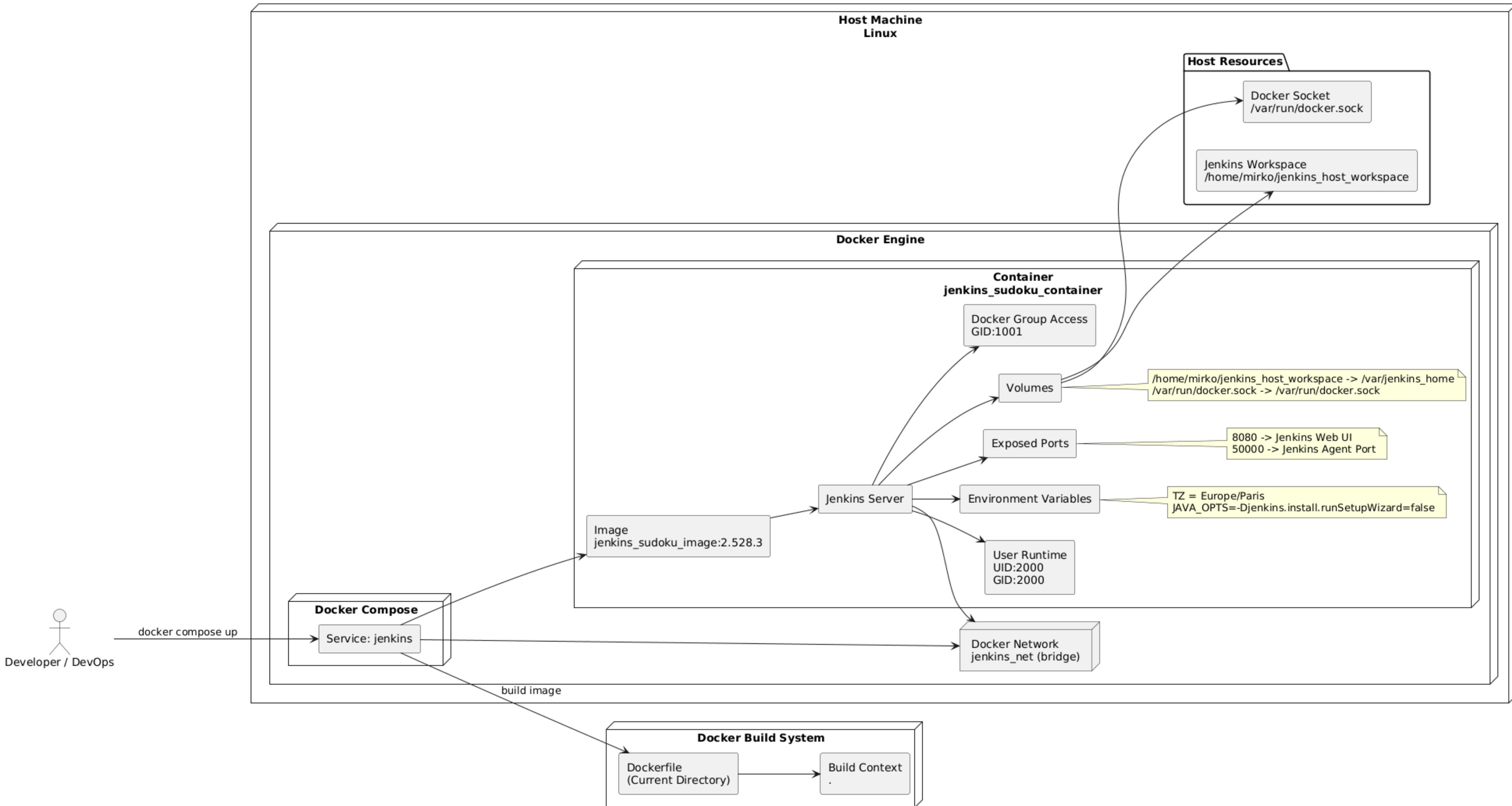
Flutter Project Setup Script



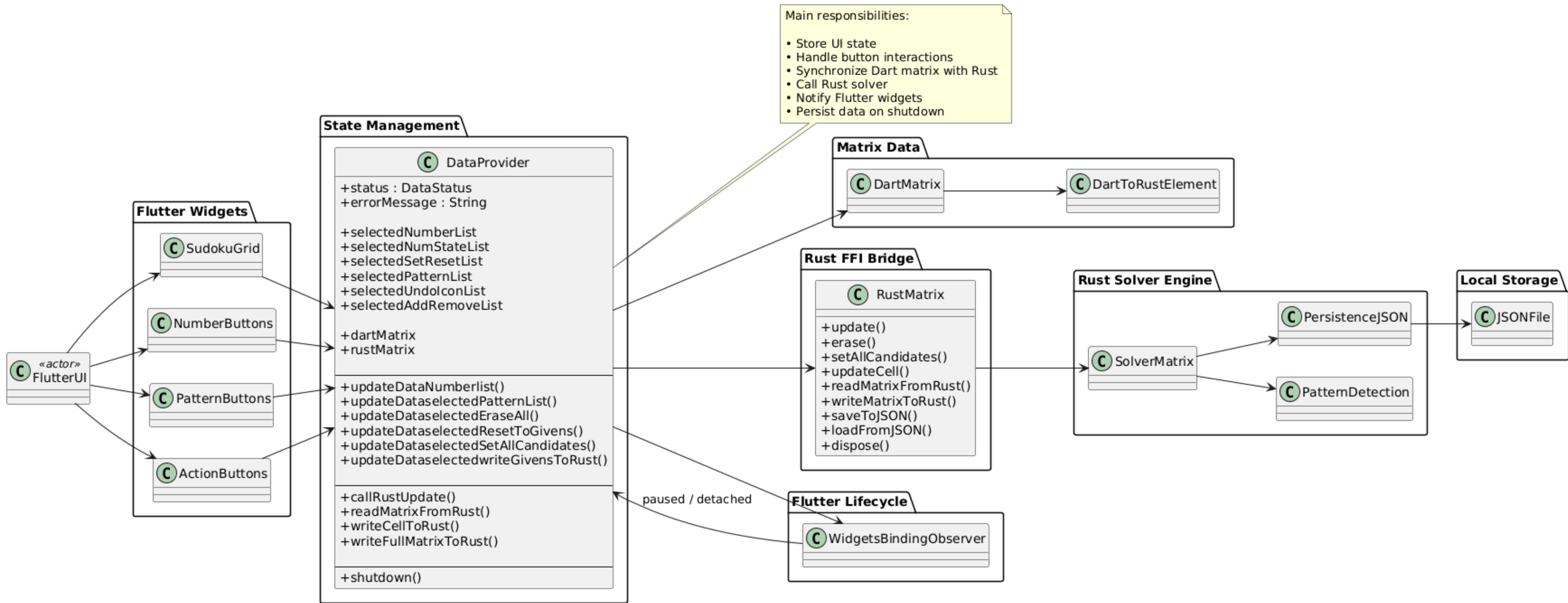
Flutter & Gradle Cache Reset Script



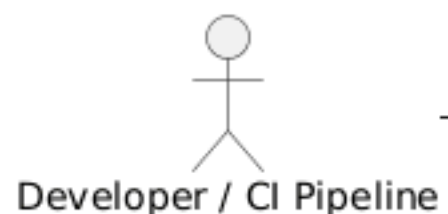
Jenkins Sudoku CI - Docker Compose Deployment



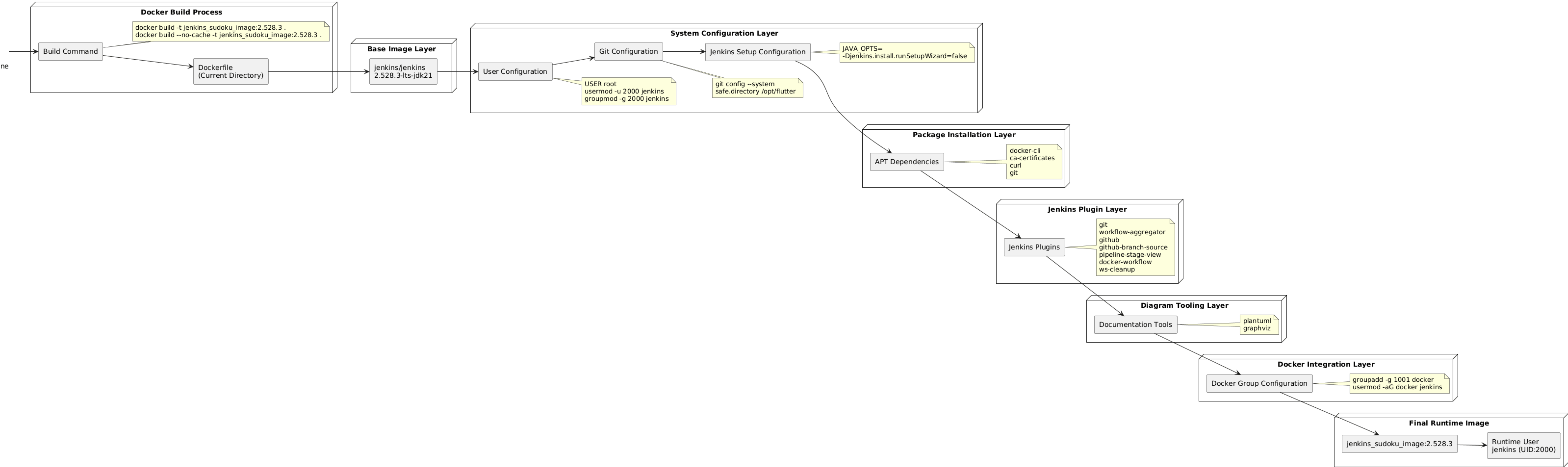
Flutter DataProvider - State Management & Rust FFI Bridge



Jenkins Sudoku CI - Docker Image Build



Developer / CI Pipeline



Dart Export Module

utils

C shared_types

C help_page

C user_feedback

C size_config

C sudoku_logging

home

C AddRemoveToggle

C CustomAppBar

C SecureButton

C TopActionButtons

providers

C DataProvider

rust_ffi

C RustMatrix

app

C MyHomePage

C MyApp

sudoku

C SudokuGrid

C SudokuBlock

C SudokuElement

sudoku_buttons

C ToggleButtonsSample

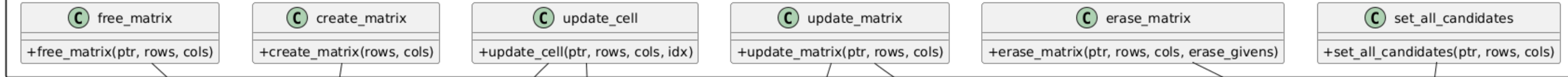
C NumberButtons

C PatternButtons

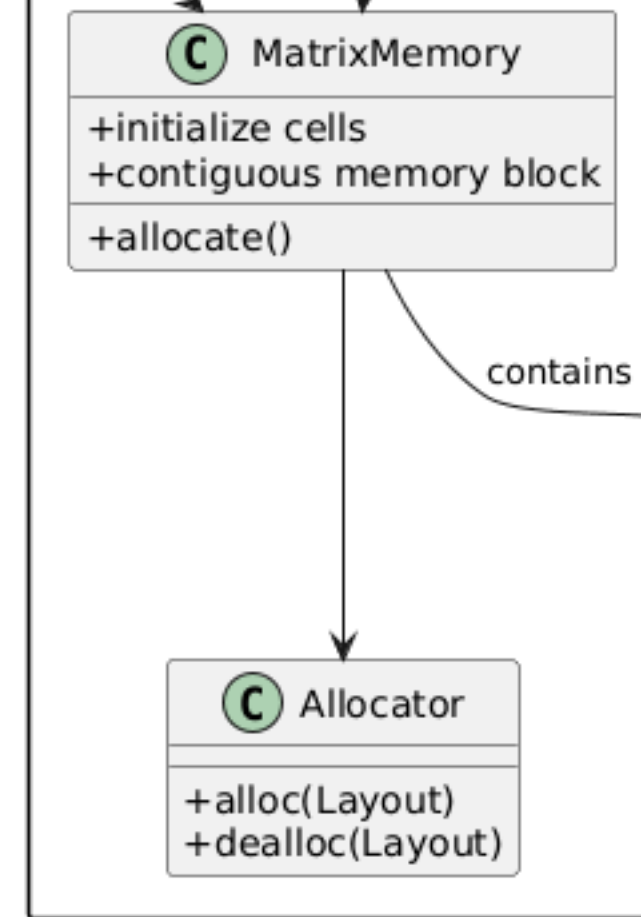
C SetResetButtons

Rust Sudoku Matrix FFI Architecture

C FFI Interface



Memory Model



Pattern Detection Engine

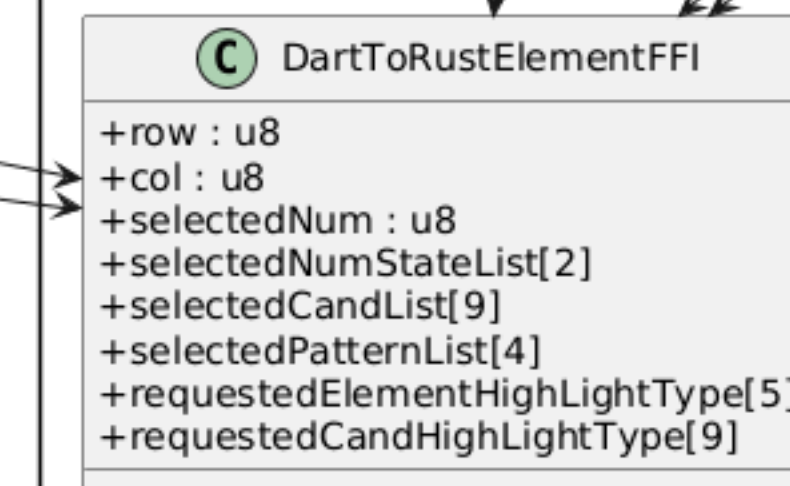


Detects patterns such as:

- pairs
- singles
- highlight candidates

Runs solver analysis over entire matrix

FFI Data Model

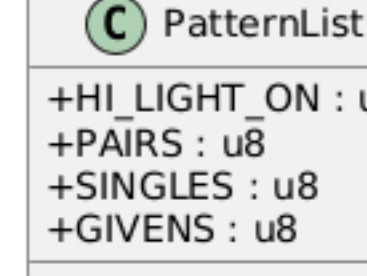
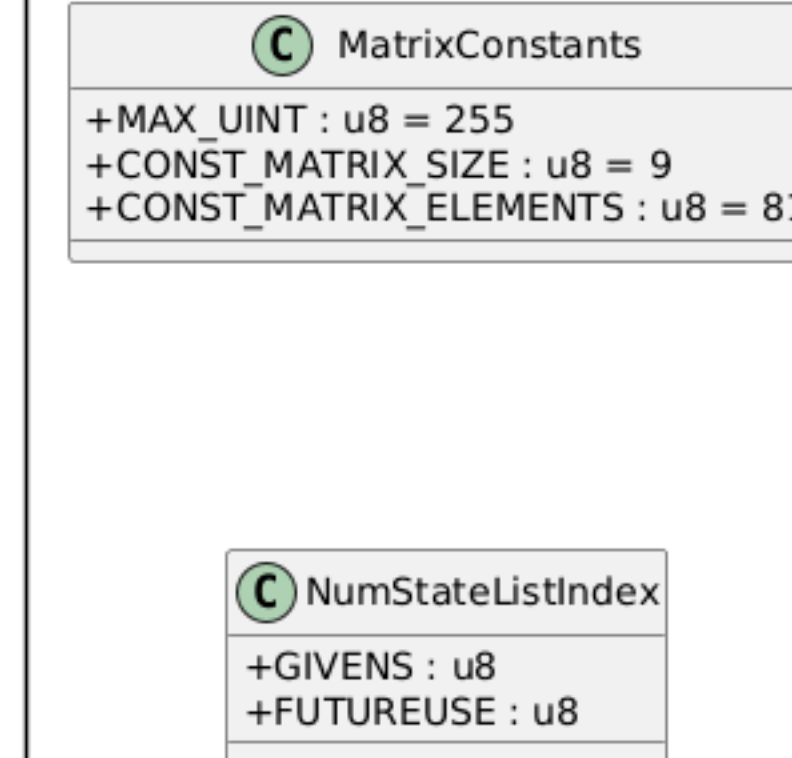


Represents a single Sudoku cell.

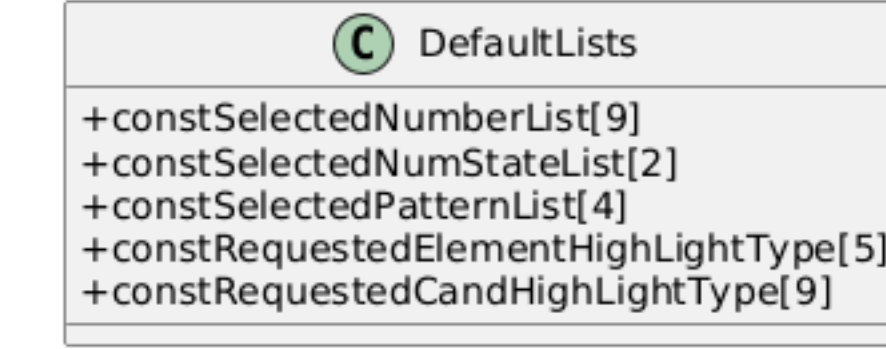
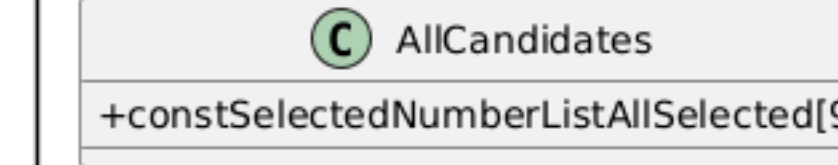
Used for FFI communication between Dart and Rust.

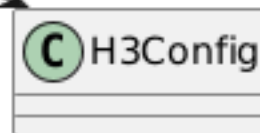
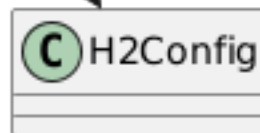
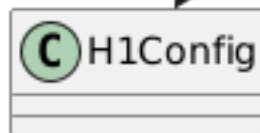
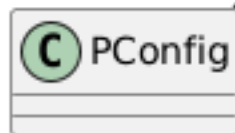
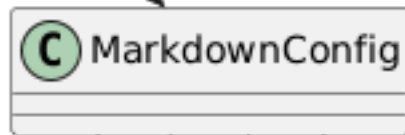
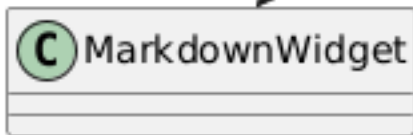
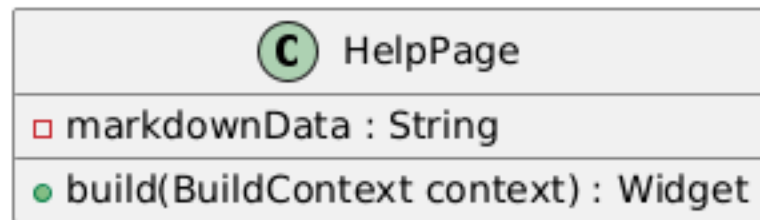
Total cells = 81
Matrix size = 9 x 9

Constants

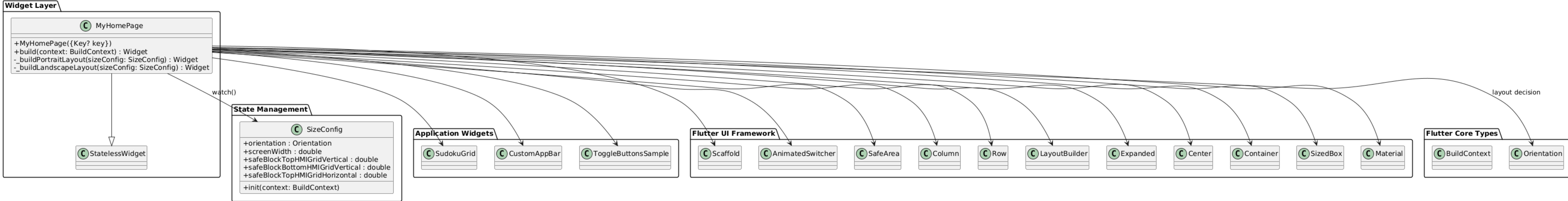


Default Data Arrays

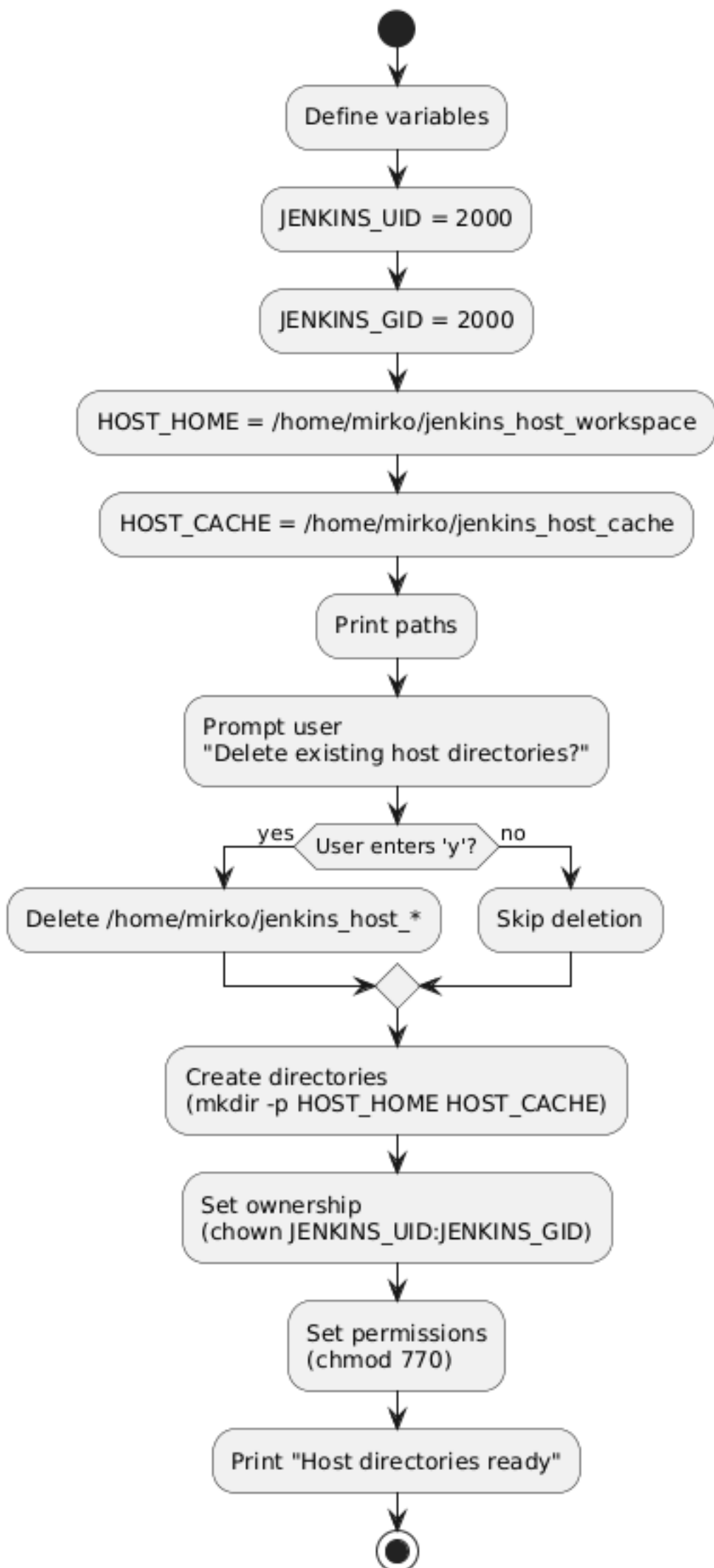




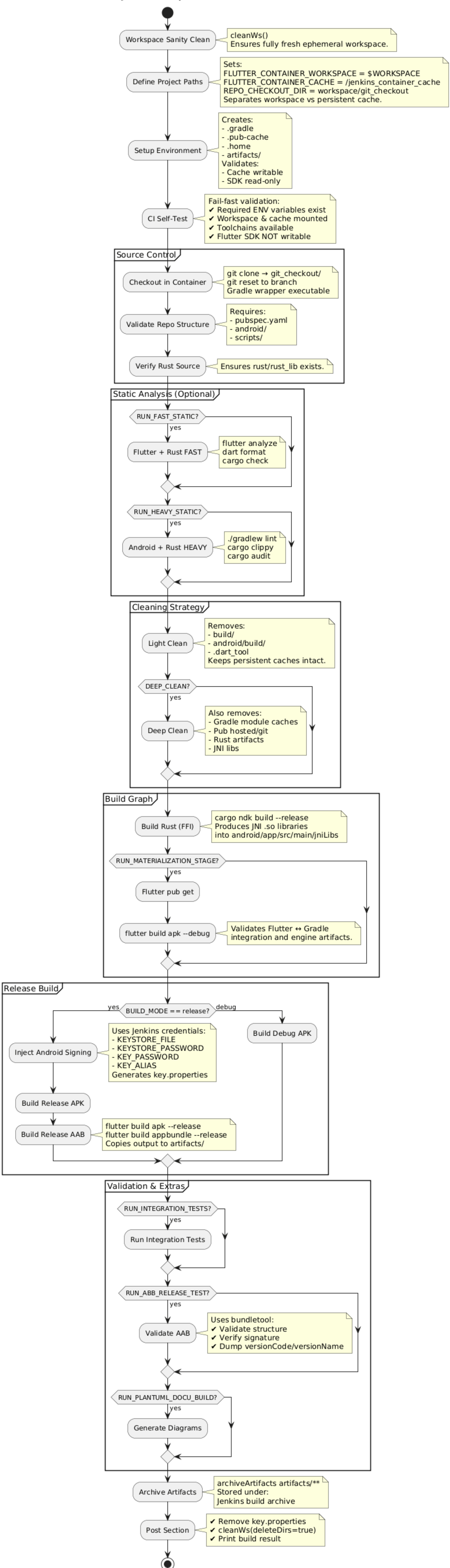
Flutter Widget Class Diagram - MyHomePage



Jenkins Host Directory Setup Script



Jenkins CI Pipeline - Flutter + Rust + Android



```
graph TD; VSCode[VSCode] --> FlutterCLI[FlutterCLI]; VSCode --> Gradle[Gradle]; VSCode --> Docker[Docker]; VSCode --> PlantUML[PlantUML];
```

VSCode

FlutterCLI

Gradle

Docker

PlantUML

Rust Sudoku Solver - Crate Module Structure

crate root (lib.rs)

lib.rs

Entry point of the Rust crate.

Defines internal modules:

- ffi
- process_data
- store_data

Re-exports the public API:
`pub use ffi::*;`

Dart / Flutter App

FFI calls

mod

mod

mod

ffi.rs
(FFI interface)

Provides C-compatible API
for Dart/Flutter via FFI.

Functions:

- create_matrix
- update_cell
- update_matrix
- erase_matrix
- set_all_candidates
- free_matrix

uses

uses

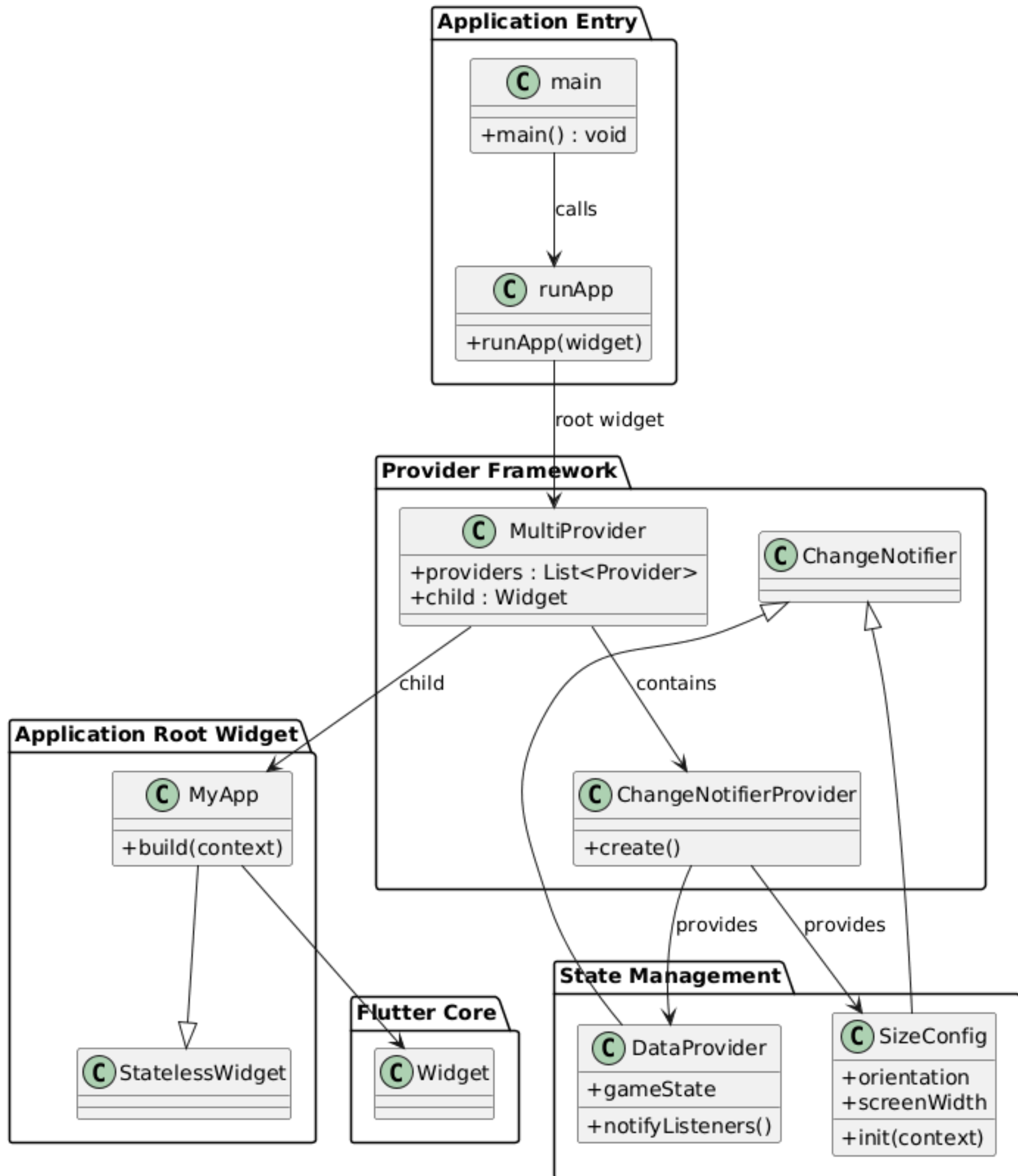
process_data.rs
(Sudoku solver logic)

store_data.rs
(JSON persistence)

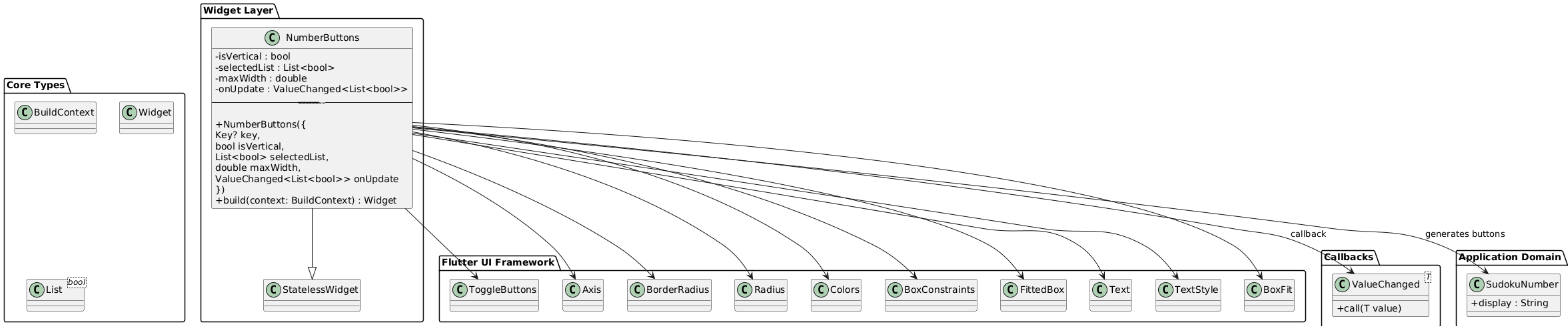
Sudoku solving logic:
- check_one_element()
- check_all_elements()

Handles persistence:
- load matrix
- save matrix
- JSON serialization

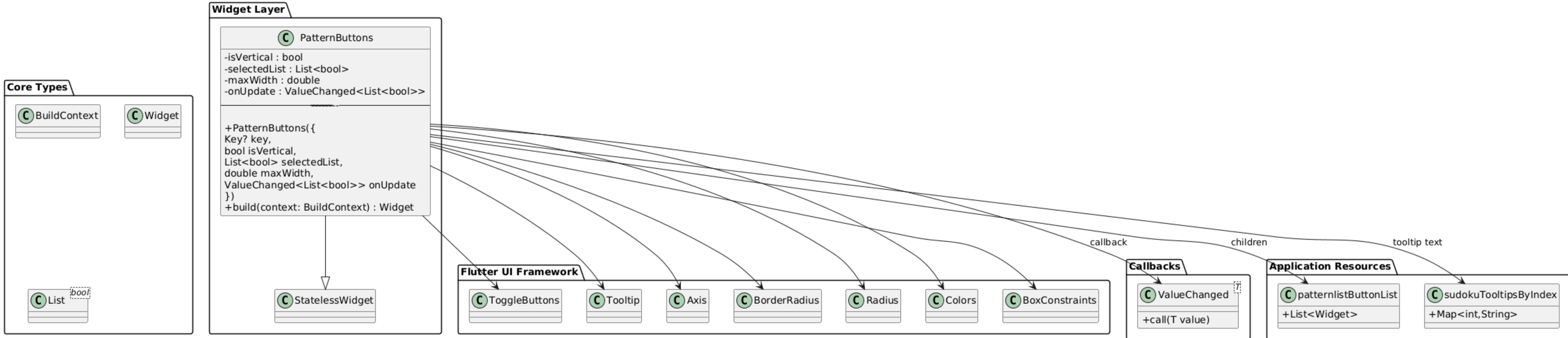
Flutter App Bootstrap & Provider Architecture



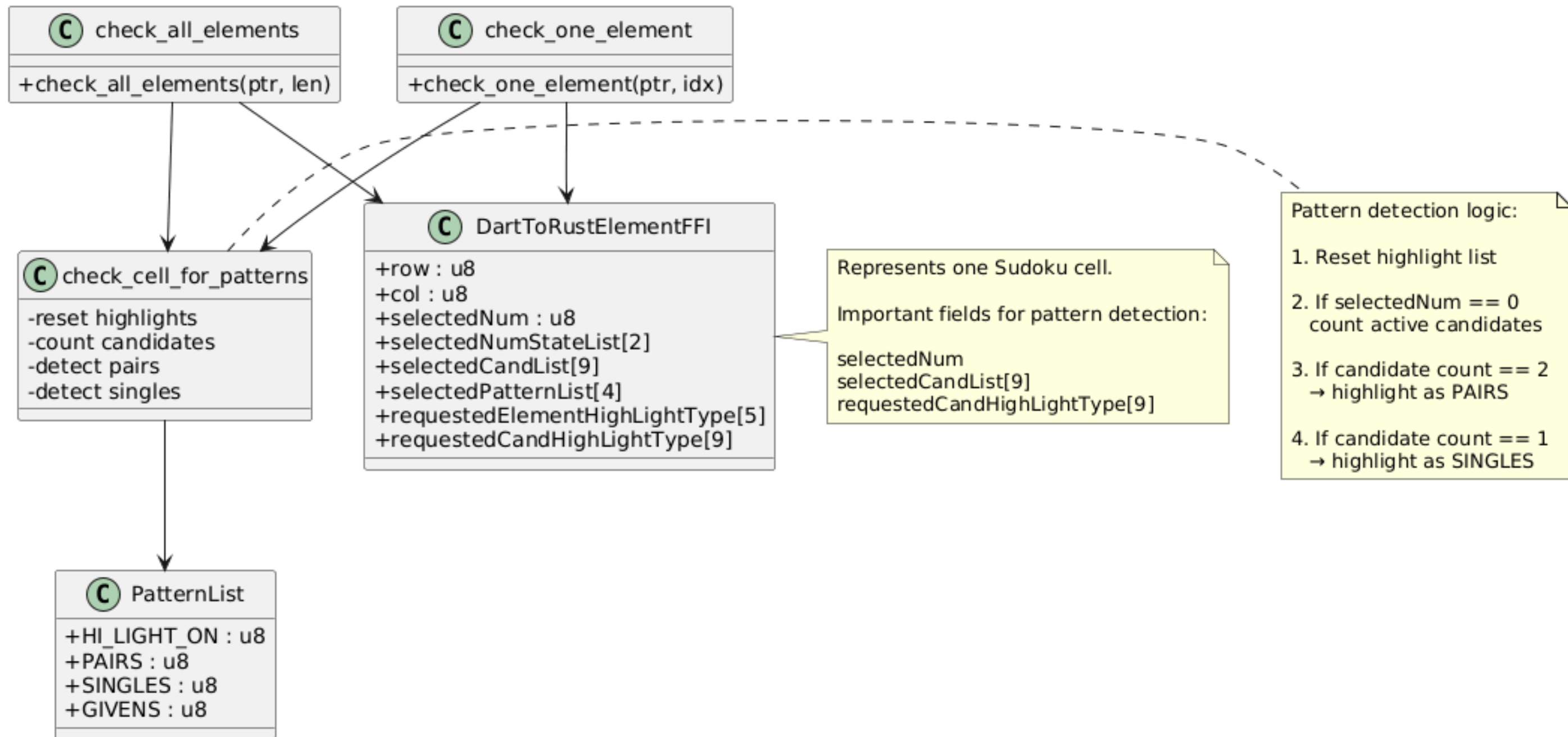
Flutter Widget Class Diagram - NumberButtons



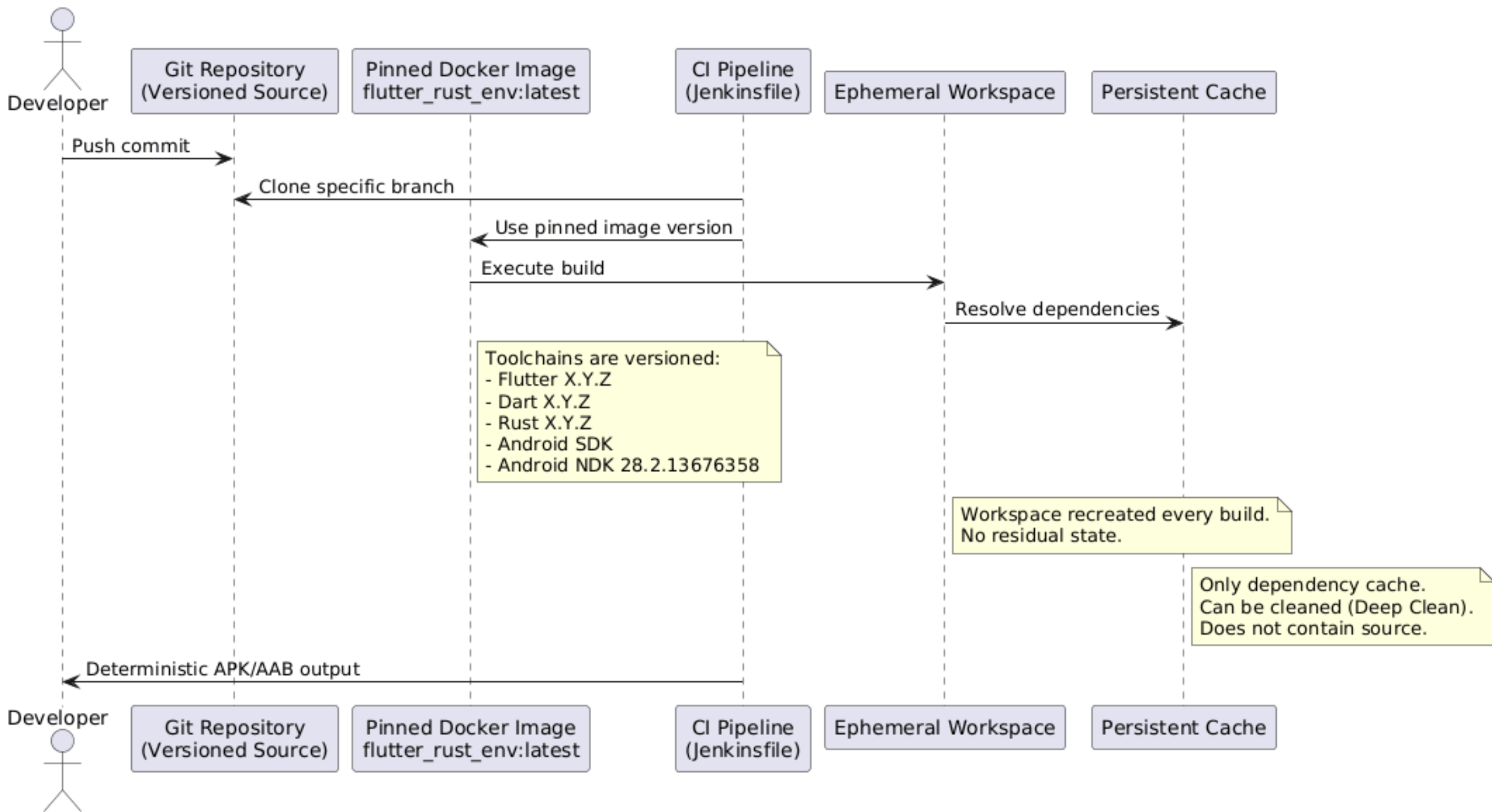
Flutter Widget Class Diagram - PatternButtons



Sudoku Pattern Detection Engine (process_data.rs)

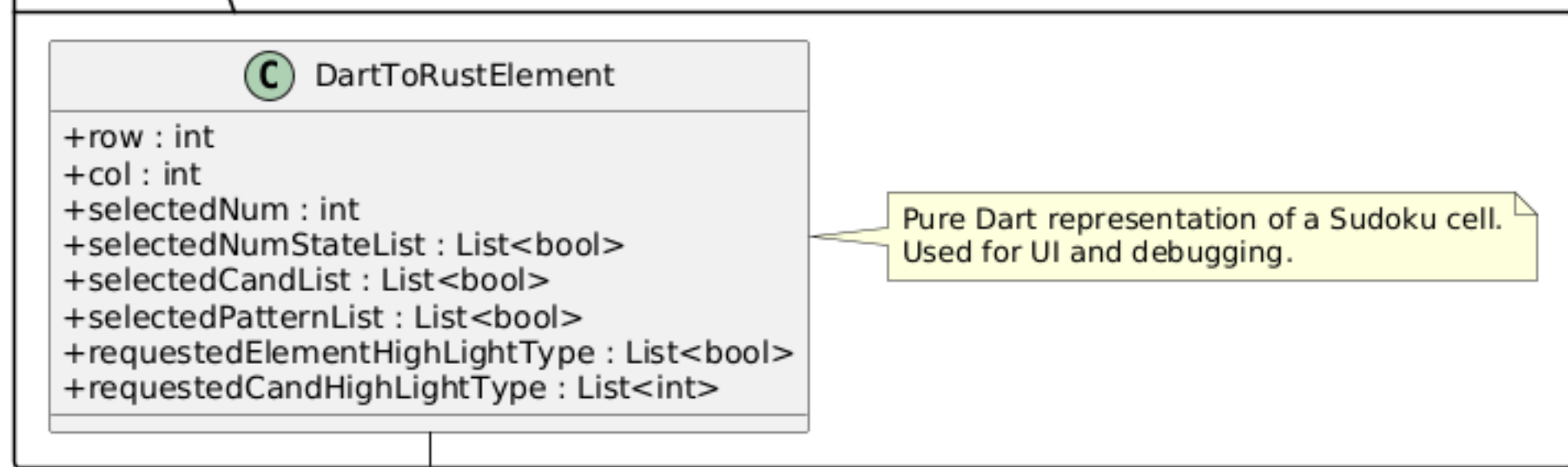


Deterministic Build Model

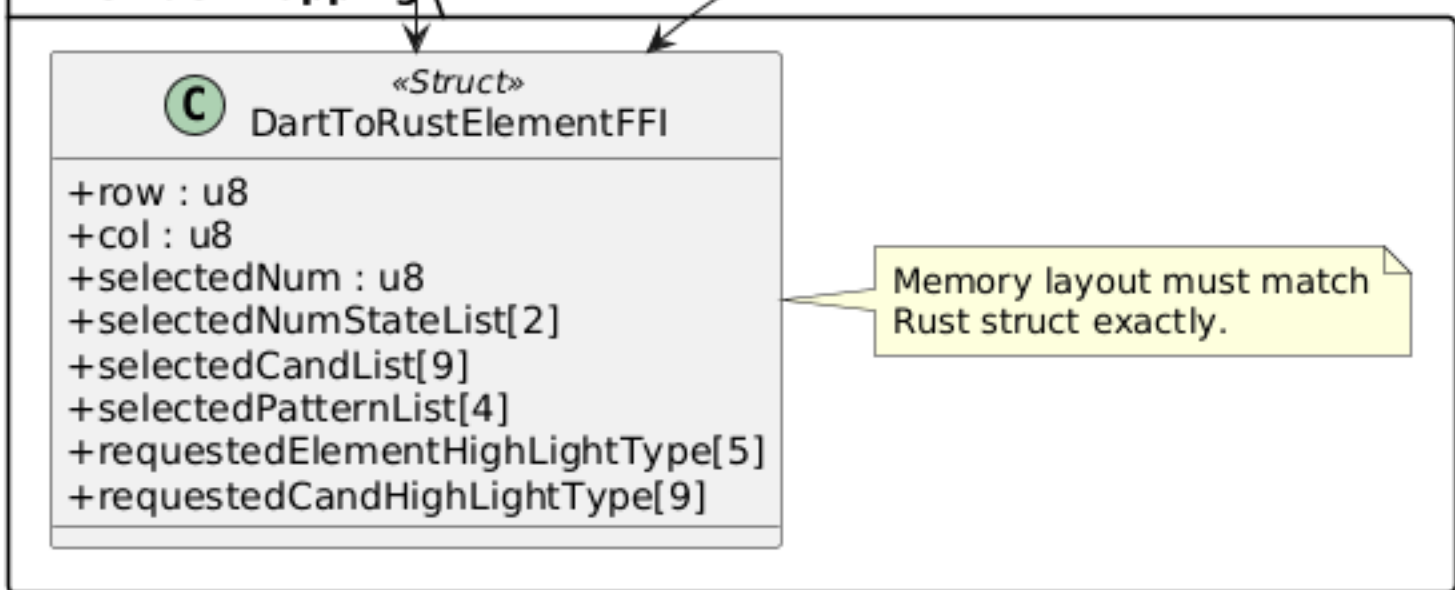


Dart ↔ Rust FFI Bridge (rust_matrix.dart)

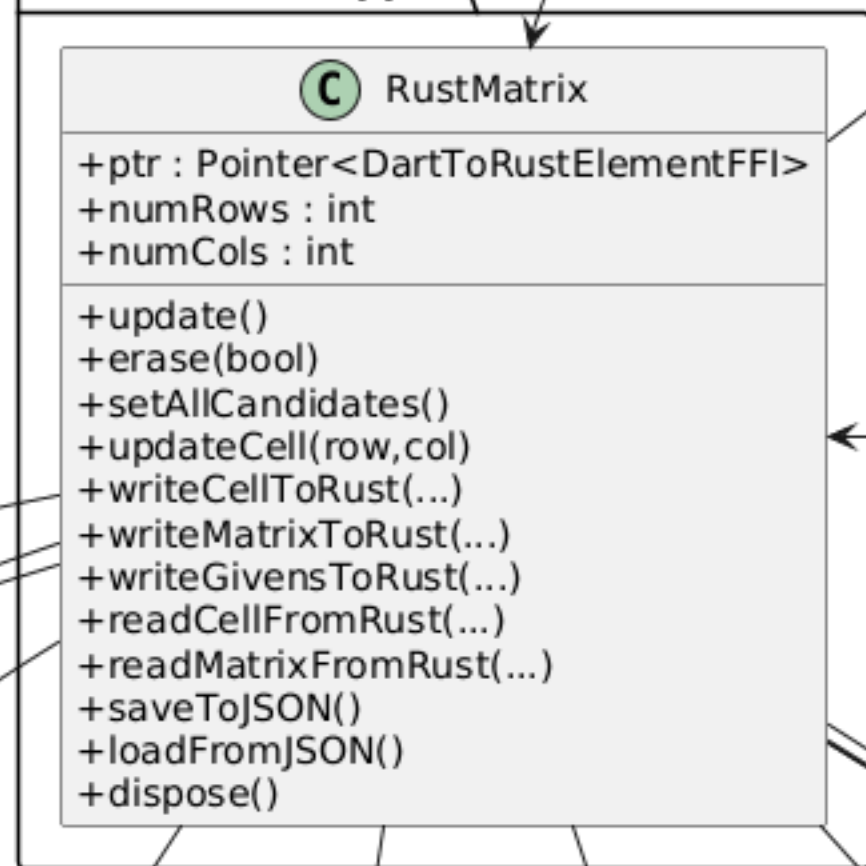
Dart Model



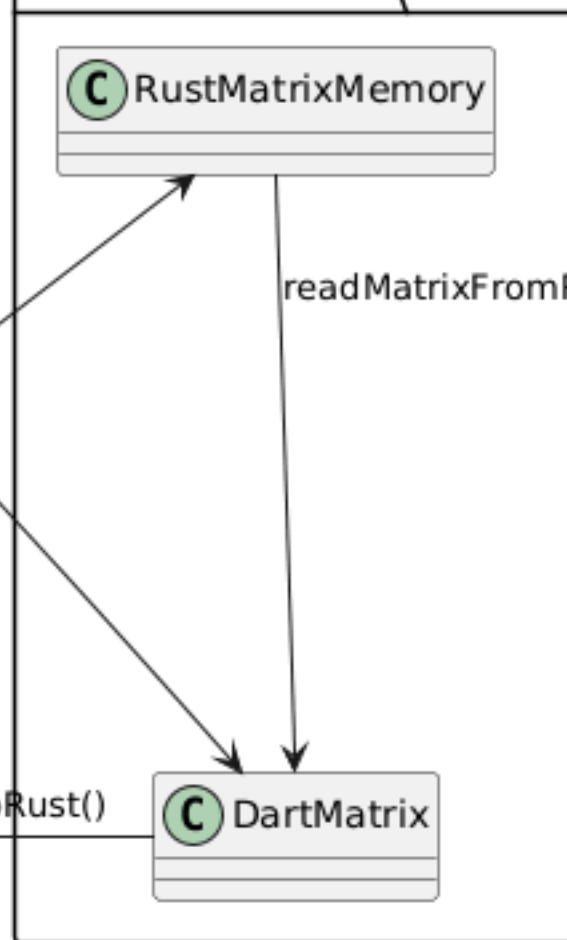
FFI Struct Mapping



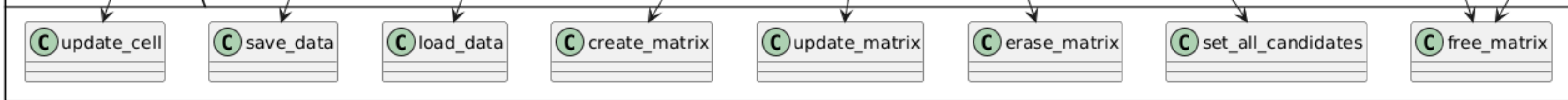
RustMatrix Wrapper



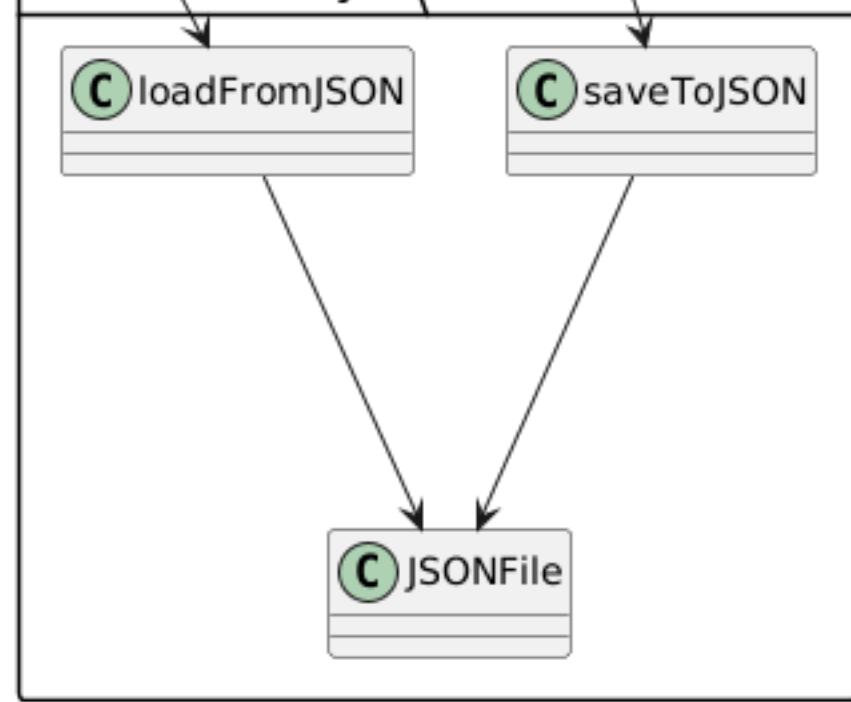
Matrix Data Flow



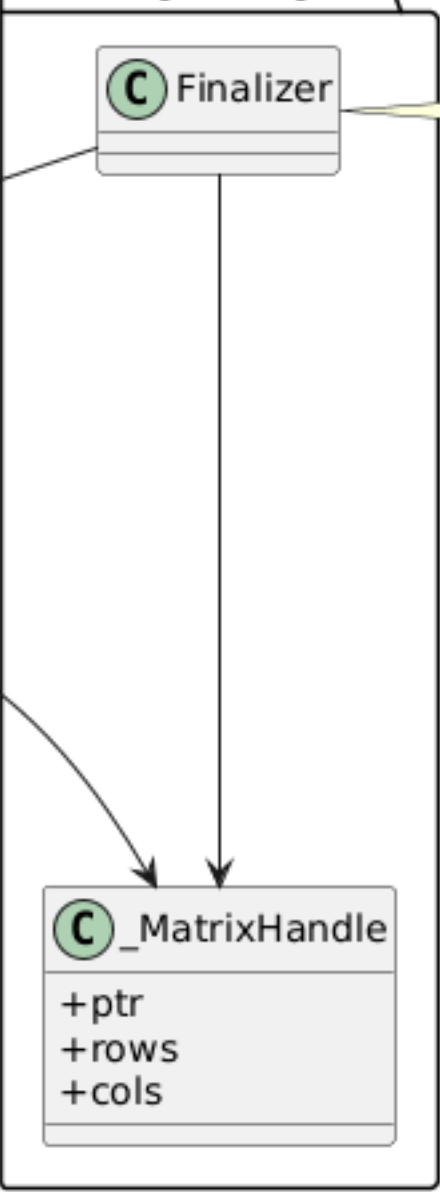
Rust FFI Functions



Persistence Layer



Memory Lifecycle



When Dart GC collects RustMatrix, Finalizer automatically calls free_matrix(ptr).

copied to / from

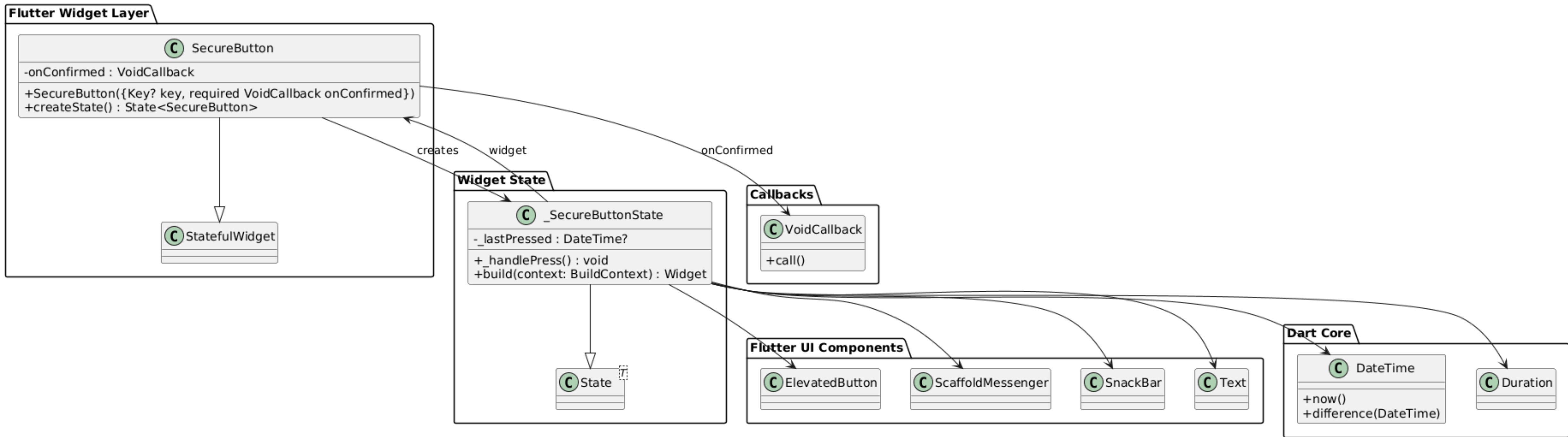
owns pointer

create / update

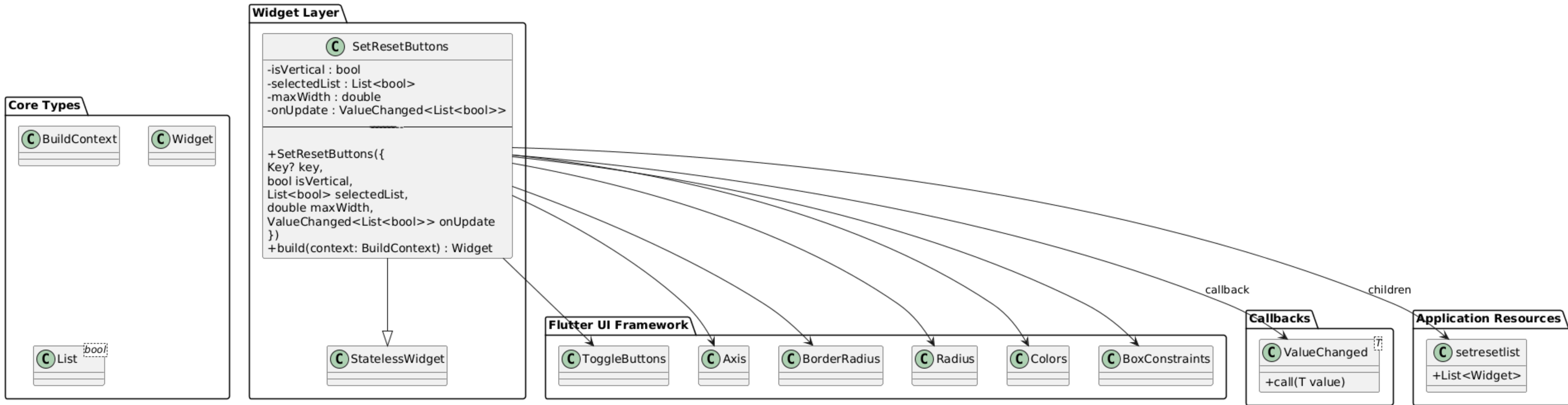
readMatrixFromRust()

writeMatrixToRust()

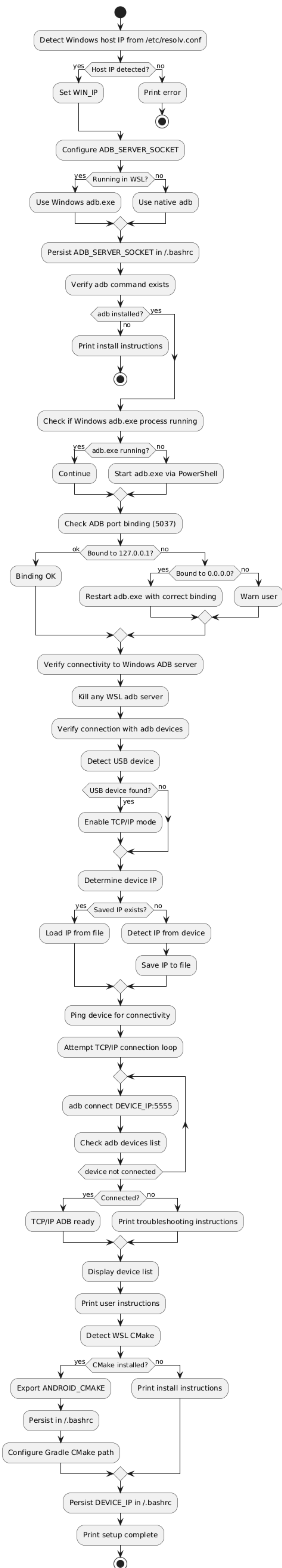
Flutter Widget Class Diagram - SecureButton



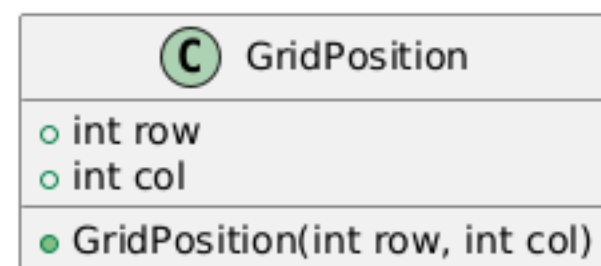
Flutter Widget Class Diagram - SetResetButtons



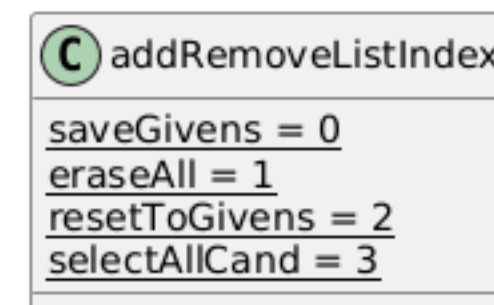
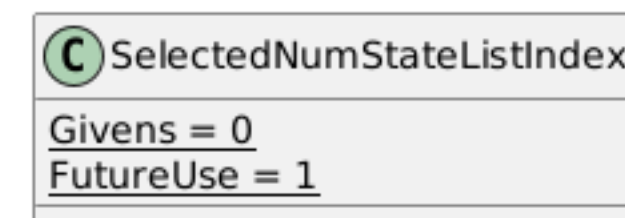
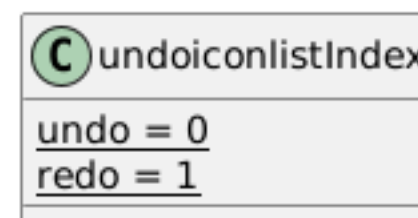
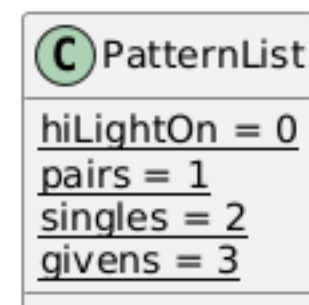
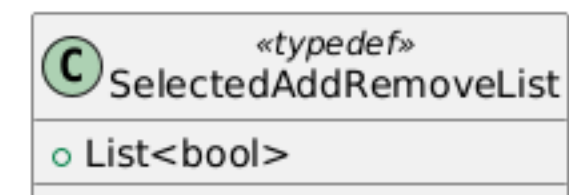
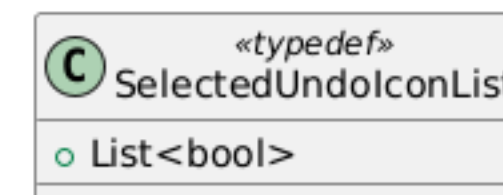
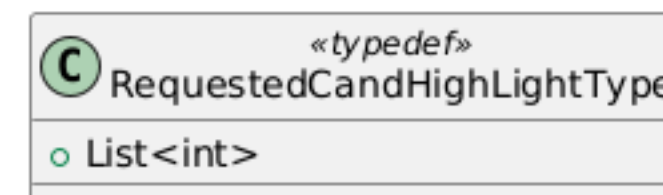
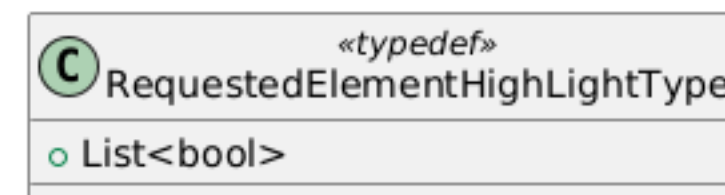
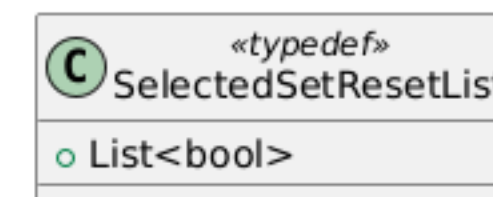
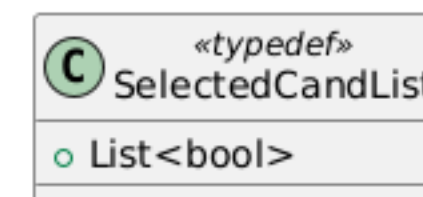
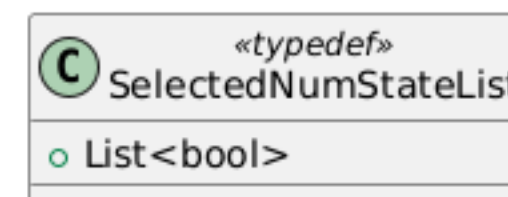
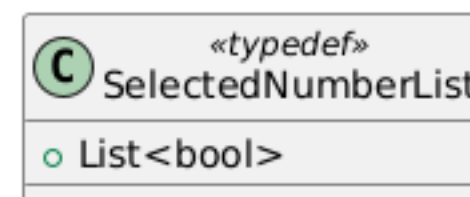
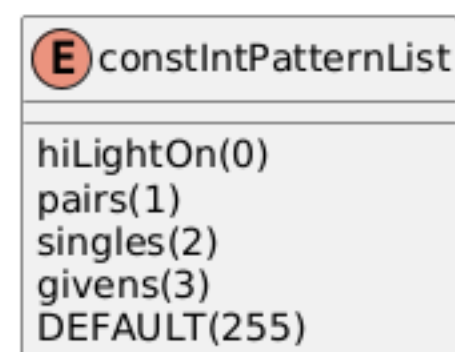
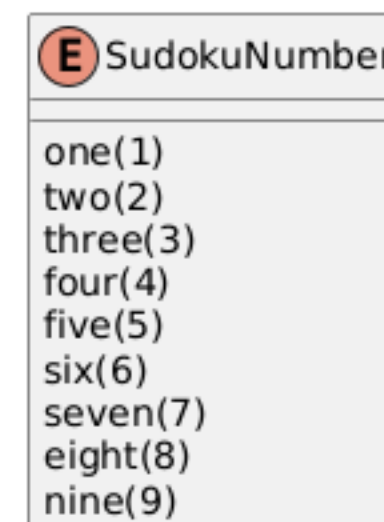
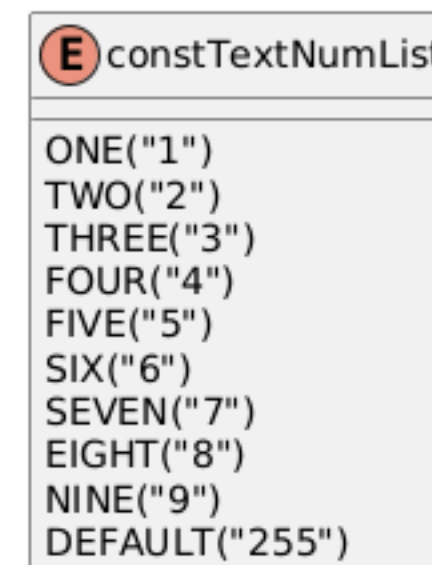
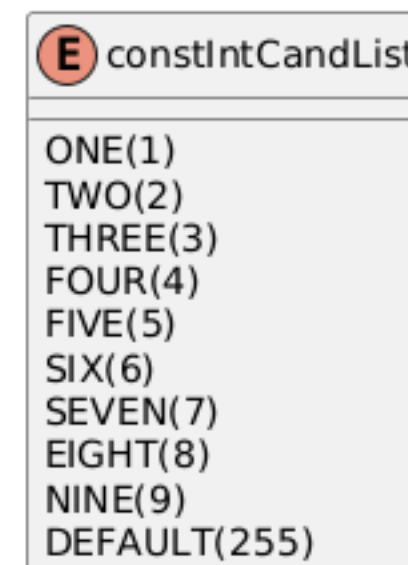
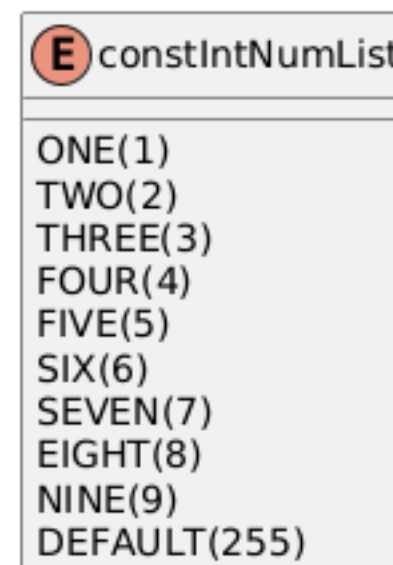
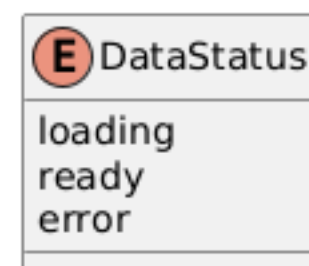
WSL ADB over TCP/IP Setup Workflow



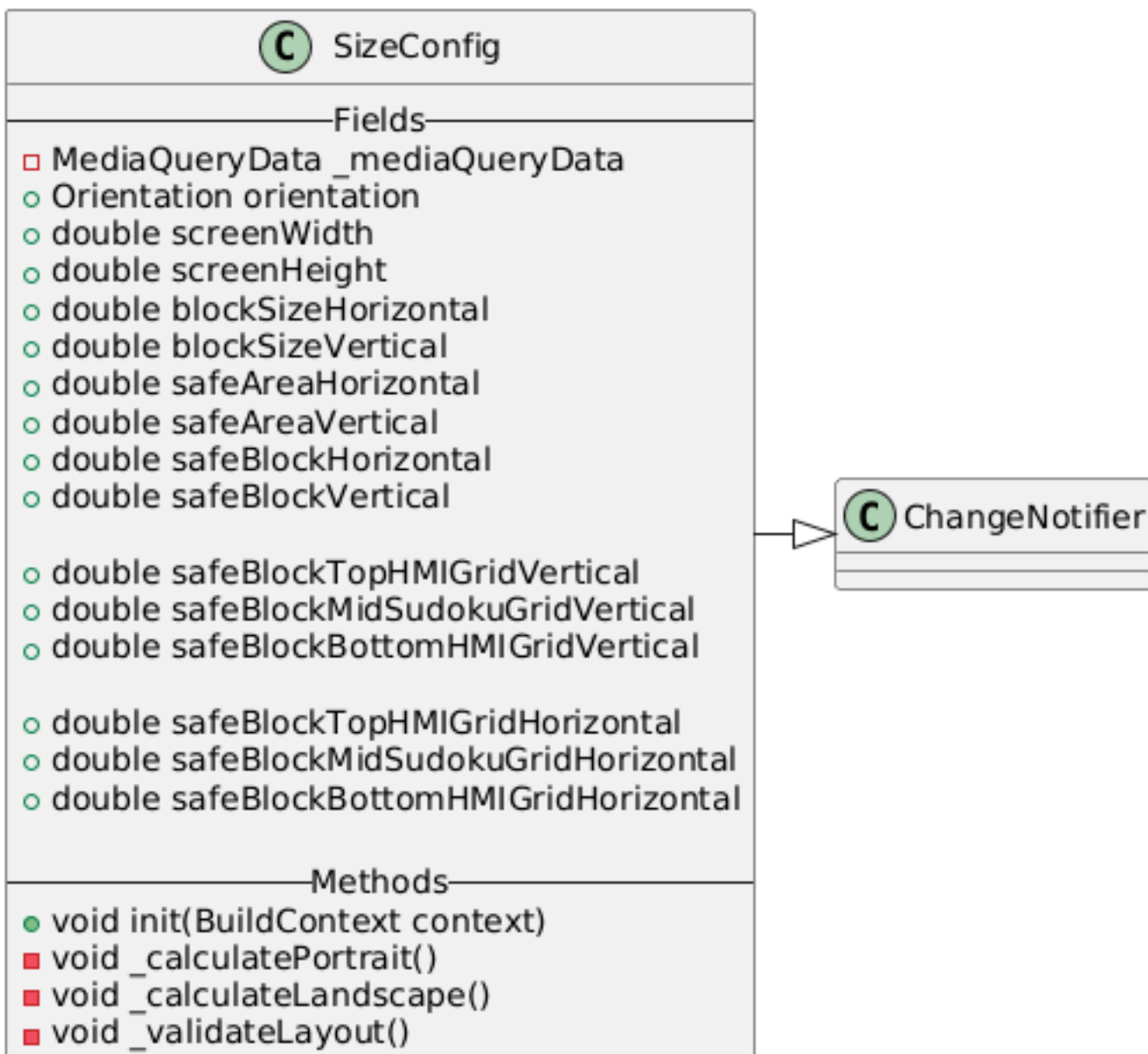
Shared Types, Enums, Constants, and Utility Classes



getRowColFromId(int id, int numColumns)
 - Computes global Sudoku coordinates
 - Based on 3x3 block subdivision



SizeConfig Class Diagram



Sudoku Data Persistence (store_data.rs)

JSON Data Model

C AppData

+rows : u8
+cols : u8
+elements : Vec_SerializableElement

C SerializableElement

+row : u8
+col : u8
+selectedNum : u8
+selectedNumStateList : Vec_u8
+selectedCandList : Vec_u8
+selectedPatternList : Vec_u8
+requestedElementHighLightType : Vec_u8
+requestedCandHighLightType : Vec_u8

JSON-friendly version of the Sudoku cell.
Rust Vec<u8> arrays are used instead of fixed-size arrays.

Conversion Layer

C ToFFI

+convert_to_ffl()

C FromFFI

+convert_from_ffl()

FFI Memory Model

C DartToRustElementFFI

+row : u8
+col : u8
+selectedNum : u8
+selectedNumStateList[2]
+selectedCandList[9]
+selectedPatternList[4]
+requestedElementHighLightType[5]
+requestedCandHighLightType[9]

Raw struct used by Dart ↔ Rust FFI.
Uses fixed-size arrays to ensure predictable memory layout.

C Dart

FFI Persistence API

C load_data

+load_data(ptr, rows, cols, path)

C save_data

+save_data(ptr, rows, cols, path)

File System / JSON

C serde_json

C filesystem

deserialize

serialize

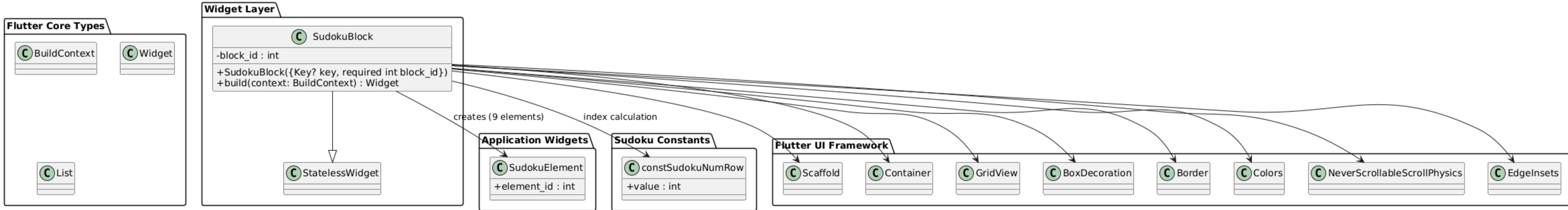
read file

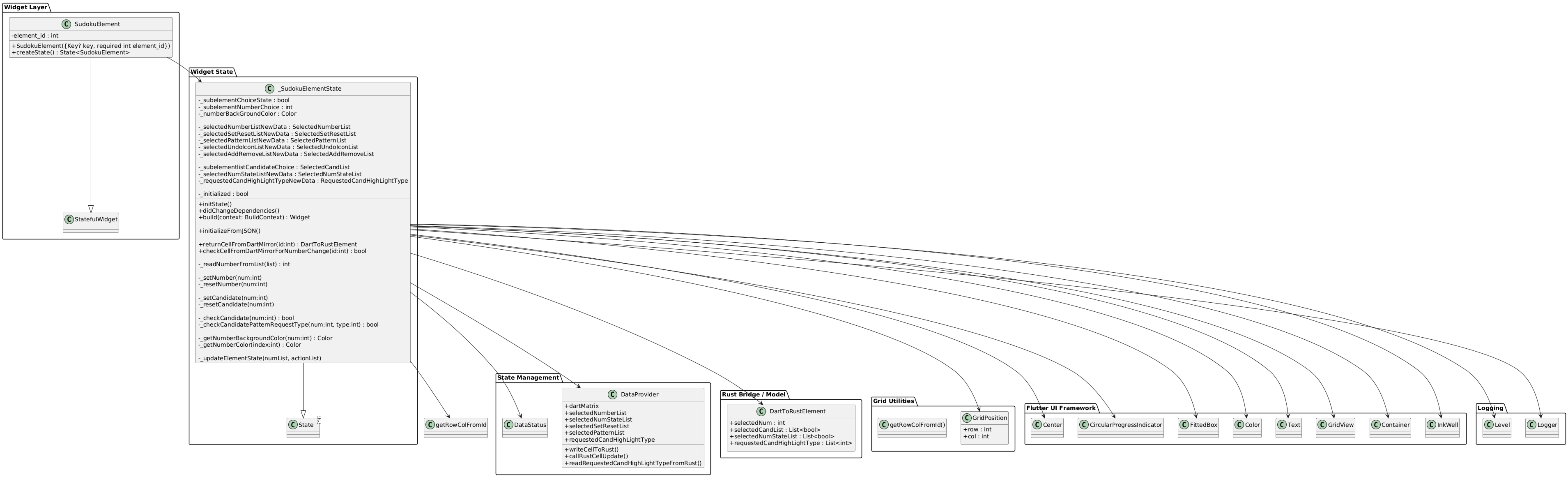
write file

contains

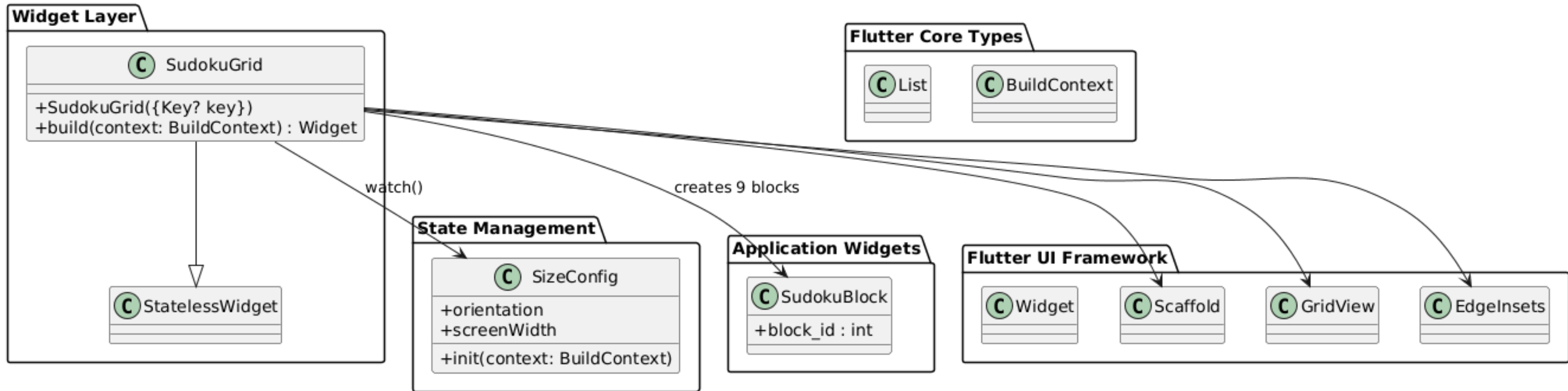
conversion

Flutter Widget Class Diagram - SudokuBlock





Flutter Widget Class Diagram - SudokuGrid



Initializes global logging
for debugging and diagnostics.

Debug build:

Logger.root.level = Level.ALL

Release build:

Logger.root.level = Level.SEVERE

Example usage:

log.info("Matrix loaded")

log.warning("Unexpected state")

log.severe("Rust call failed")

Other options :

FINEST

FINER

FINE

CONFIG

INFO

WARNING

SEVERE

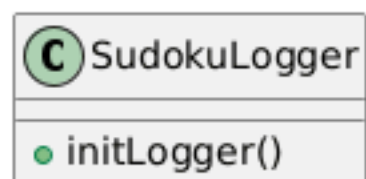
SHOUT

Log visibility:

info -> Debug: shown, Release: hidden

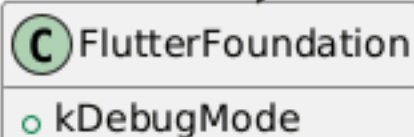
warning -> Debug: shown, Release: hidden

severe -> Debug: shown, Release: shown



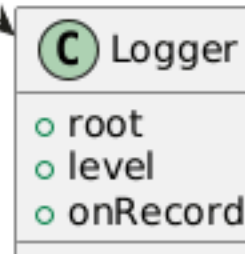
reads

configures

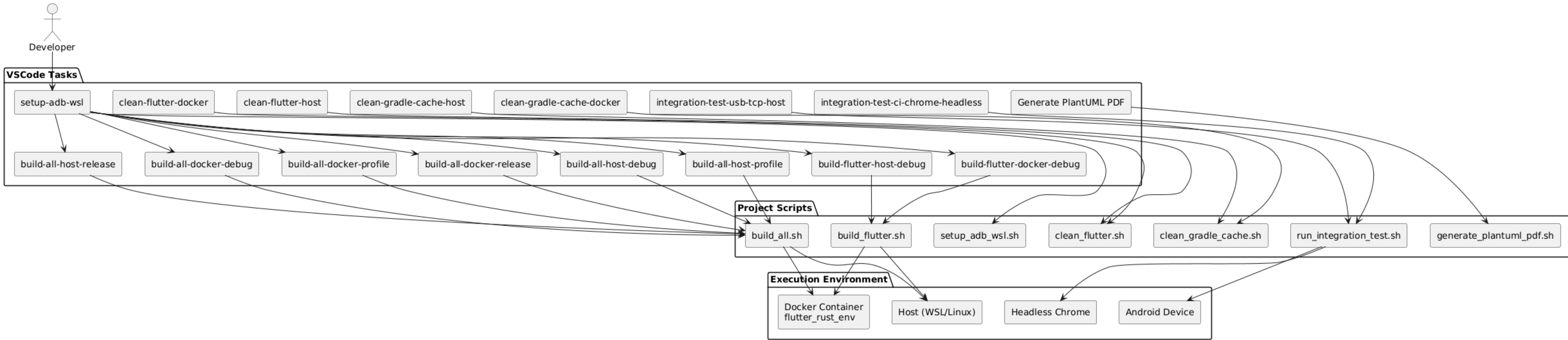


Provided by Flutter.

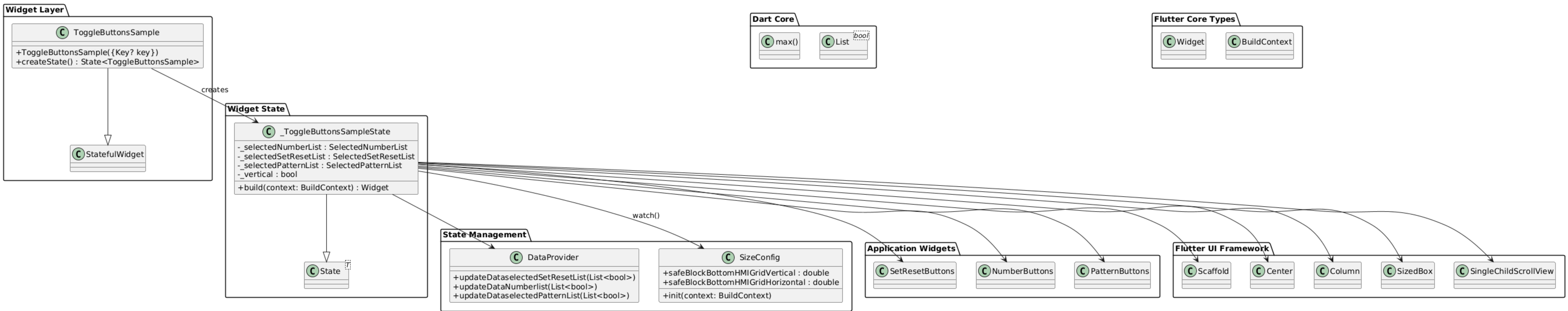
kDebugMode determines
whether the application
runs in debug or release mode.



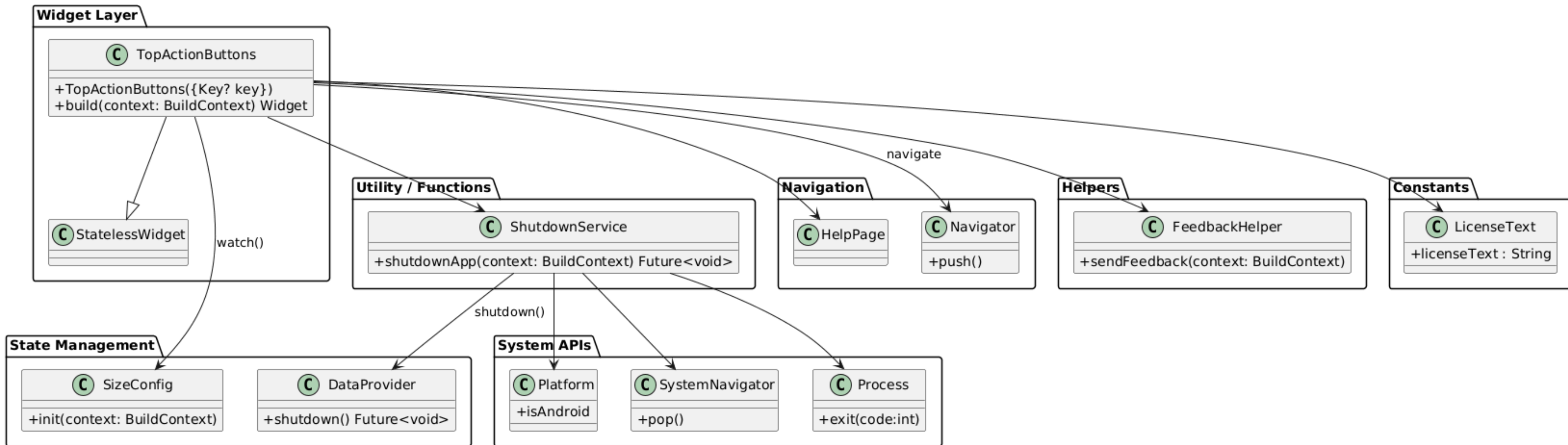
VSCode Tasks - Flutter + Rust Build Pipeline



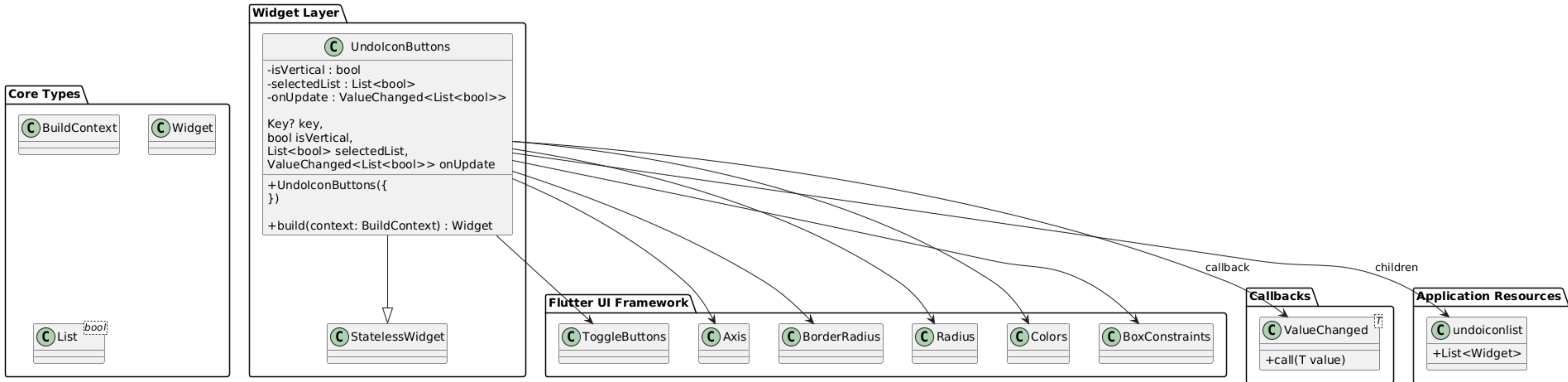
Flutter Widget Class Diagram - ToggleButtonsSample



Flutter Widget Class Diagram - TopActionButton



Flutter Widget Class Diagram - UndolconButtons



C FeedbackHelper

● sendFeedback(BuildContext)

Launches the email client
to send user feedback.

C Uri

C url_launcher

Workspace vs Cache Security Model

